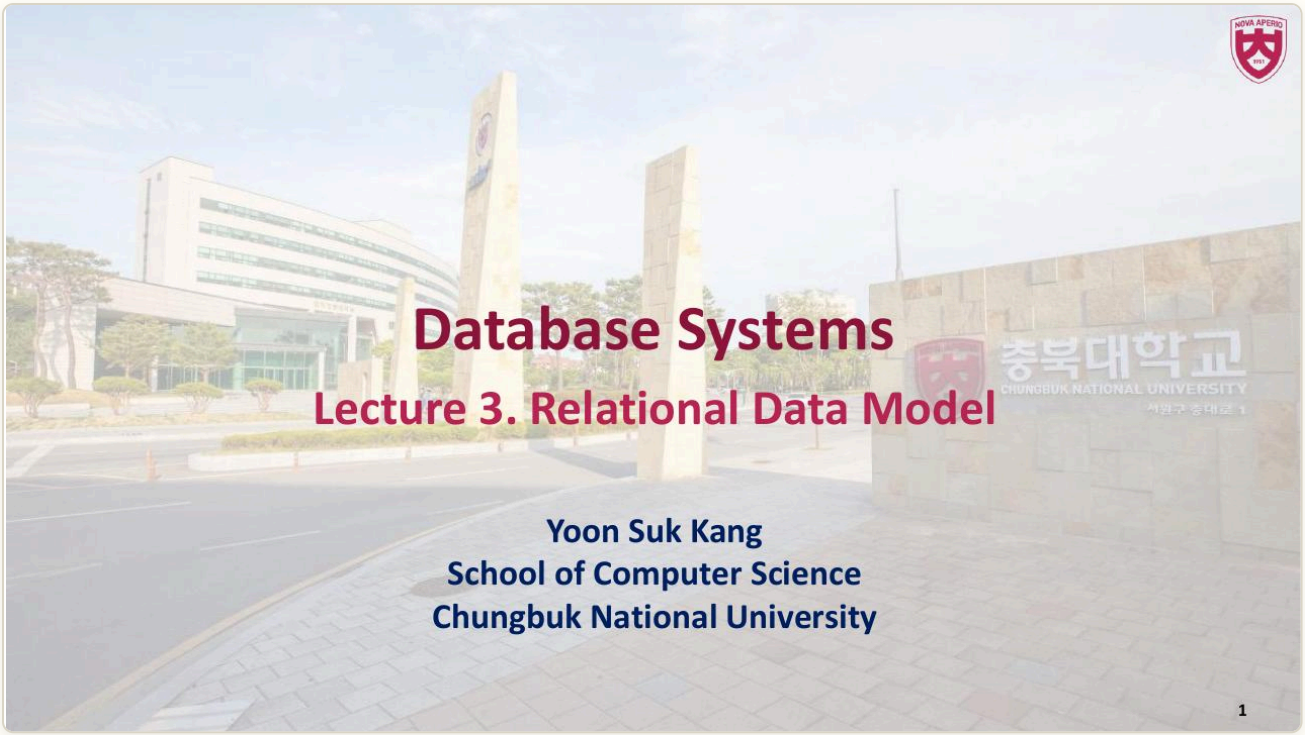
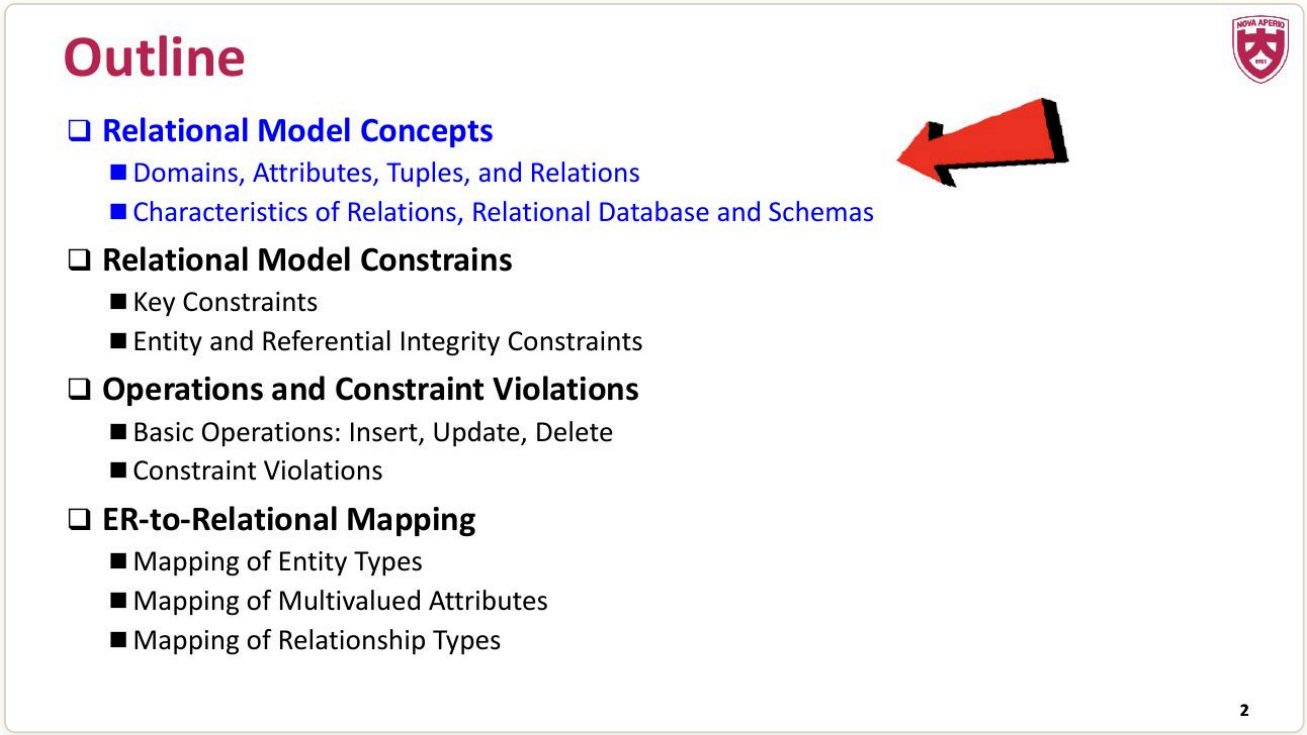


데이터베이스 시스템 — 3강: 관계형 데이터 모델

- Yoon Suk Kang · 충북대 소프트웨어학부

필기 ✍️

💡 이해 이 강의: ① 관계형 모델 용어 ② 제약 3개 ③ 연산별 위반 ④ ER → 표로 바꾸기(매핑)



목차 (Outline)

- **관계형 모델 개념** — 도메인·속성·튜플·릴레이션 / 릴레이션의 특성, 관계형 DB와 스키마
- **관계형 모델 제약** — 키 제약 / 개체·참조 무결성
- **연산과 제약 위반** — Insert·Update·Delete / 제약 위반
- **ER → 관계 매핑** — 엔티티 타입 / 다중값 속성 / 관계 타입

필기 ✍️

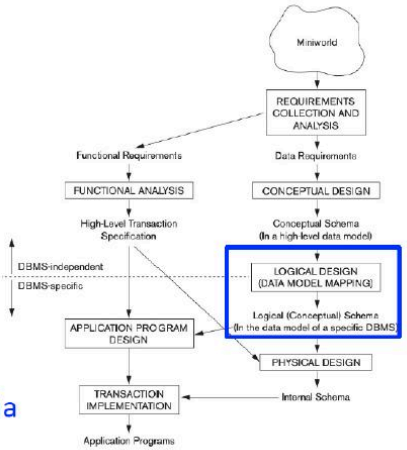
★ 시험 시험 핵심은 제약 3개 + 매핑 규칙. 연습문제 1~4가 그대로 나오기 좋음

Recap: Database Design Process

❑ **Phase 1: Requirements Collection and Analysis**
- ■ Characterize the data needs of the prospective DB users
- ■ Collect and analyze data requirements for the database
  - ❑ **Output**: a set of users' detailed requirements

❑ **Phase 2: Conceptual Design**
- ■ Choosing a data model
- ■ Translating the requirements into a conceptual schema

❑ **Phase 3: Implementation**
- ■ Implementing the DB based on the conceptual DB schema
  - ❑ Logical design: the database schema
  - ❑ Physical design: the physical layout of the DB



3

복습: DB 설계 과정

- **1단계 요구사항 수집·분석** — 결과: 사용자 요구사항 명세
- **2단계 개념 설계** — 데이터 모델 고르기, 요구사항 → 개념 스키마(ER)
- **3단계 구현** — 논리 설계(DB 스키마) / 물리 설계(물리 배치)

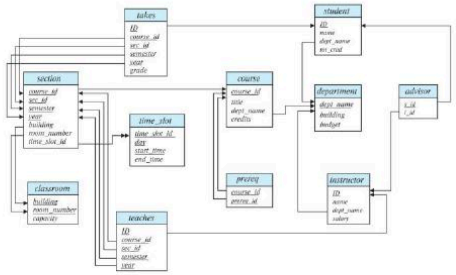
필기 ✍️

💡 이해 2강이 2단계(ER), 이번 3강의 매핑이 3단계 '논리 설계'

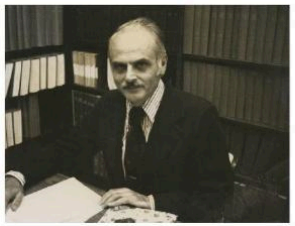
Relational Data Model

❑ What is a Relational Data Model?

- Database design simplifies data representation, making data management efficient
  - ❑ Developed by Ted Codd of IBM in 1970
- Using ***relations*** (i.e., *tables*) to represent data, where each relation consists of ***tuples*** (i.e., *rows*) and ***attributes*** (i.e., *column headers*)



Example of Relational Databases



Ted Codd of IBM Research in 1970

4

관계형 데이터 모델

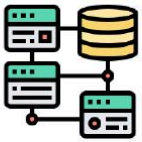
- 관계형 데이터 모델이란?
  - 데이터 표현을 단순하게 해 관리를 효율적으로
  - **1970년 IBM의 Ted Codd**가 개발
  - **relation(= 표)**으로 데이터를 나타냄 — 각 relation은 **tuple(= 행)**과 **attribute(= 열 머리)**로 구성

필기 ✍️

★ 시험 동어 짝: relation = table, tuple = row, attribute = column header

Advantages of Relational Model

- ❑ **Simplicity**
  - ■ Using a flat table structure, making it easy to understand and manage
- ❑ **Data Integrity**
  - ■ Ensuring *the data in the database is valid, consistent, and accurate*
- ❑ **Easy Data Manipulation**
  - ■ Supporting *powerful, high-level data manipulation languages such as SQL*
    - ❑ Users can flexibly manipulate data along with complex conditions
- ❑ **Security**
  - ■ Offering robust security features,
    - ❑ E.g., user authentication, authorization mechanisms, data encryption, and so on



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관계형 모델의 장점

- **Simplicity** — 평평한 표 구조라 이해·관리가 쉬움
- **Data Integrity** — DB 데이터가 유효하고 일관되고 정확하도록
- **Easy Data Manipulation** — SQL 같은 강력한 고수준 언어, 복잡한 조건도 유연하게
- **Security** — 사용자 인증, 권한, 데이터 암호화 등

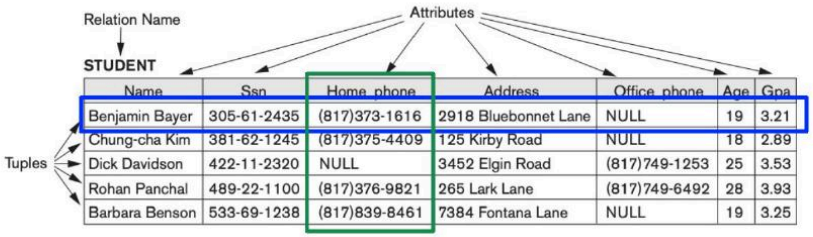
필기 ✍️

★ 시험 강점 4개: Simplicity · Data Integrity · Easy Data Manipulation · Security

How To Represent a Database?

❑ **Represent a Database as a Collection of *Relations* (i.e., *Tables*)**

- ■ A relation consists of a set of ***tuples*** (i.e., *rows*)
  - ❑ Each *row* in the table represents a collection of related data values
  - ❑ A row represents a fact that typically corresponds to a real-world entity or relationship
- ■ A ***column header*** represents an *attribute* defining a property of each object



Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

Each row represents facts about a particular student entity (object)
The column headers (i.e., attributes) specify how to interpret the data values in each row.
All values in a column are of the same data type

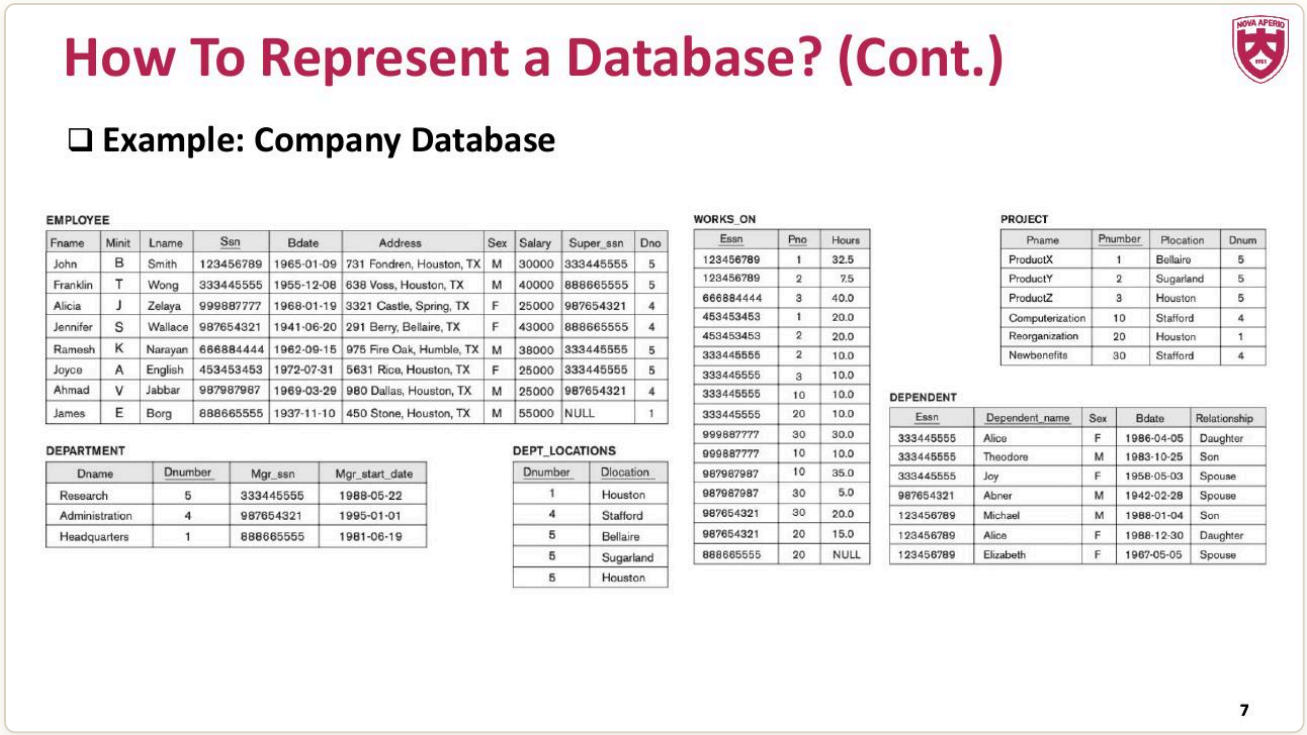
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DB를 어떻게 표현할까?

- DB를 **relation(표)들의 모음**으로 나타낸다
  - relation은 **tuple(행)**의 집합
    - 행 하나 = 관련 데이터 값의 모음
    - 행 하나 = 실세계 엔티티나 관계에 해당하는 **사실(fact)**
  - **열 머리(column header)** = 각 객체의 성질을 정하는 **attribute**
- 그림: STUDENT — 행은 학생 한 명의 사실, 열 머리는 값을 해석하는 법, 한 열의 값은 모두 같은 자료형

필기 ✍️

💡 이해 빨간 상자 = 행(튜플), 초록 상자 = 열(속성) — 한 열 = 같은 자료형



How To Represent a Database? (Cont.)

❑ **Example: Company Database**

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

7

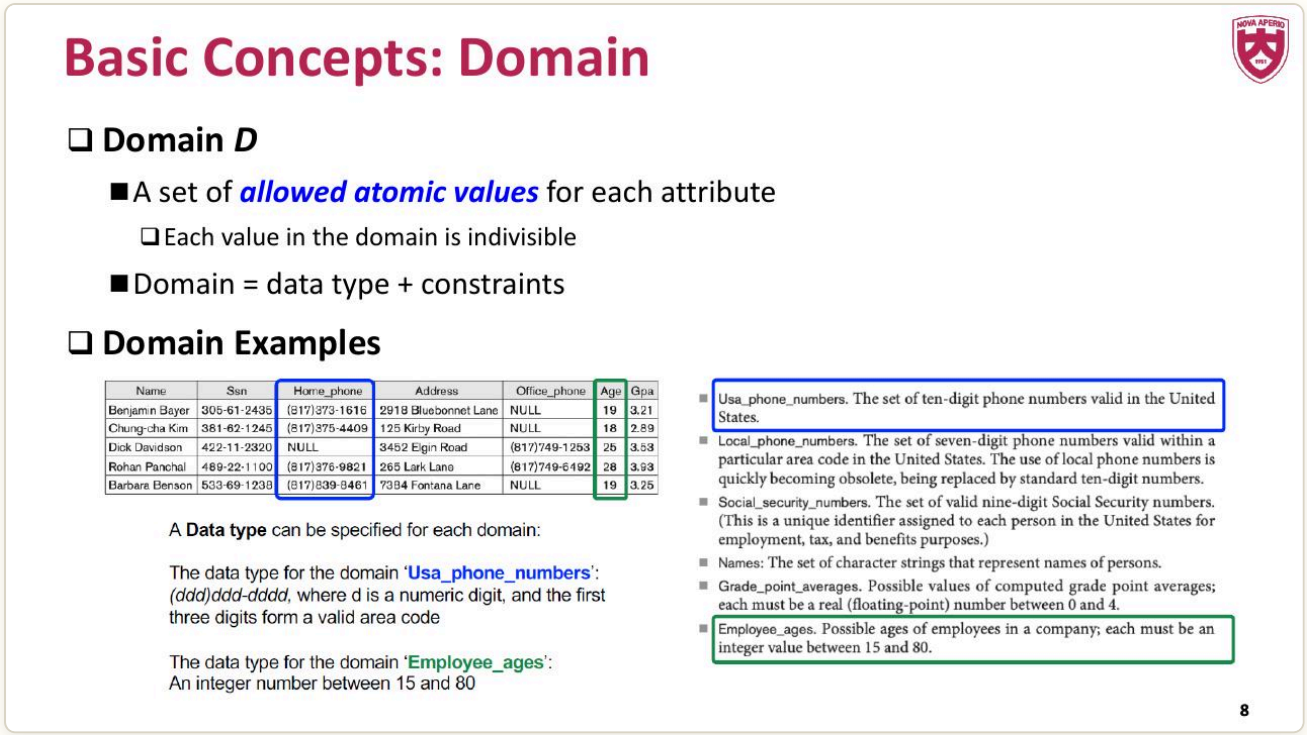
DB를 어떻게 표현할까? (예: 회사 DB)

- COMPANY DB의 표들: EMPLOYEE, DEPARTMENT, DEPT_LOCATIONS, WORKS_ON, PROJECT, DEPENDENT

필기 ✍️

★ 시험 이 회사 DB 상태가 연습문제 3(p.36~)과 SQL·관계 대수 예제에 계속 쓰인다 — 눈에 익혀 두기

✏ 예.풀이 Smith(123456789)의 Dno = 5 → Research 부서, 상사 Super_ssn = 333445555(Wong)



Basic Concepts: Domain

❑ **Domain *D***

■ A set of ***allowed atomic values*** for each attribute

❑ Each value in the domain is indivisible

■ Domain = data type + constraints

❑ **Domain Examples**

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

A **Data type** can be specified for each domain:

The data type for the domain '**Usa_phone_numbers**': *(ddd)ddd-dddd*, where d is a numeric digit, and the first three digits form a valid area code

The data type for the domain '**Employee_ages**': An integer number between 15 and 80

- Usa_phone_numbers. The set of ten-digit phone numbers valid in the United States.
- Local_phone_numbers. The set of seven-digit phone numbers valid within a particular area code in the United States. The use of local phone numbers is quickly becoming obsolete, being replaced by standard ten-digit numbers.
- Social_security_numbers. The set of valid nine-digit Social Security numbers. (This is a unique identifier assigned to each person in the United States for employment, tax, and benefits purposes.)
- Names: The set of character strings that represent names of persons.
- Grade_point_averages. Possible values of computed grade point averages; each must be a real (floating-point) number between 0 and 4.
- Employee_ages. Possible ages of employees in a company; each must be an integer value between 15 and 80.

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기본 개념: 도메인 (Domain)

- **Domain D**: 각 속성에 **허용되는 원자 값(atomic value)의 집합**
  - 도메인의 값은 **나눌 수 없다**
  - **Domain = data type + constraints**
- 예
  - Usa_phone_numbers — 미국에서 유효한 10자리 전화번호, 형식 (ddd)ddd-dddd
  - Local_phone_numbers(7자리), Social_security_numbers(9자리), Names, Grade_point_averages(0~4 실수)
  - Employee_ages — 15~80 사이 정수

필기 ✍️

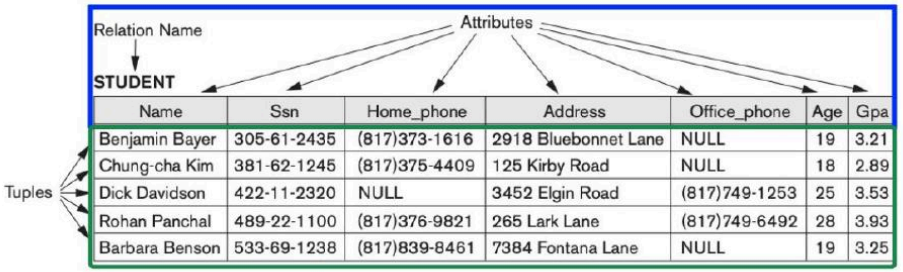
★ 시험 Domain = data type + constraints (공식처럼)

Relation Schema and Instance



❑ What is a Relation?

- Relation Schema: Relation name + name & type of each column
- Relation Instance: a table (a set of tuples)



Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

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릴레이션 스키마와 인스턴스

- 릴레이션이란?
  - **Relation Schema**: 릴레이션 이름 + 각 열의 이름과 타입 (파란 상자 = 머리)
  - **Relation Instance**: 표 = 튜플의 집합 (초록 상자 = 내용)

필기 ✍️

💡 이해 스키마 = 표의 틀(머리줄), 인스턴스 = 지금 들어 있는 행들

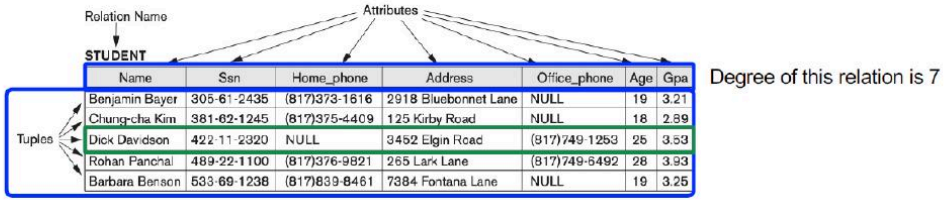
Relation Schema and Instance

❑ **Relation Schema:** $R = (A_1, \ldots, A_n)$

- ■ Made up of a **relation name** and a **list of attributes** $A_1, A_2, \ldots, A_n$
  - ❑ E.g., STUDENT = (Name, Ssn,..., Gpa)
- ■ The **degree** of a relation: the number of attributes *n* of its relation schema

❑ **Relation Instance (State):** $r(R)$

- ■ A set of *n*-tuples (i.e., rows) $r(R) = \{t_1, \ldots, t_m\}$
- ■ Tuple *t* of Relation *R:* Ordered list of *n* values $t = < v_1, \ldots, v_n >$
  - ❑ Each value v_i $(i \leq i \leq n)$ is an element of dom(A_i) or a special NULL value



Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
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Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

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릴레이션 스키마와 인스턴스 (계속)

- **Relation Schema R = (A_1, ..., A_n)**
  - 릴레이션 이름 + 속성 목록 — 예: STUDENT = (Name, Ssn, ..., Gpa)
  - **Degree(차수)**: 스키마의 속성 수 n
- **Relation Instance(State) r(R)**
  - n-tuple들의 집합 r(R) = {t_1, ..., t_m}
  - 튜플 t = < v_1, ..., v_n > — n개 값의 **순서 있는 목록**
    - 각 v_i는 dom(A_i)의 원소이거나 특별한 값 **NULL**
- 그림: 이 STUDENT의 degree는 7

필기 ✍️

★ 시험 degree = 속성(열) 수. 튜플(행) 수가 아님!
— STUDENT degree 7

⚠ 주의 ER의 관계 degree(참여 엔티티 타입 수)와 이름만 같다

Characteristics of Relations

❑ Ordering of Tuples in a Relation

- ■ A Relation is defined as a ***set*** of tuples
  - ❑ *Q. What does the order mean in a set?*
- ■ Tuples in a relation do not have any particular order
  - ❑ This is the reason why we call a relation, not a tabular

Tuple ordering 1

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

Tuple ordering 2

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21

No preference for one ordering over another
Ordering 1 is considered identical to Ordering 2

Although they are represented as a table and physically stored on disk
DBMD is not responsible for tuple ordering



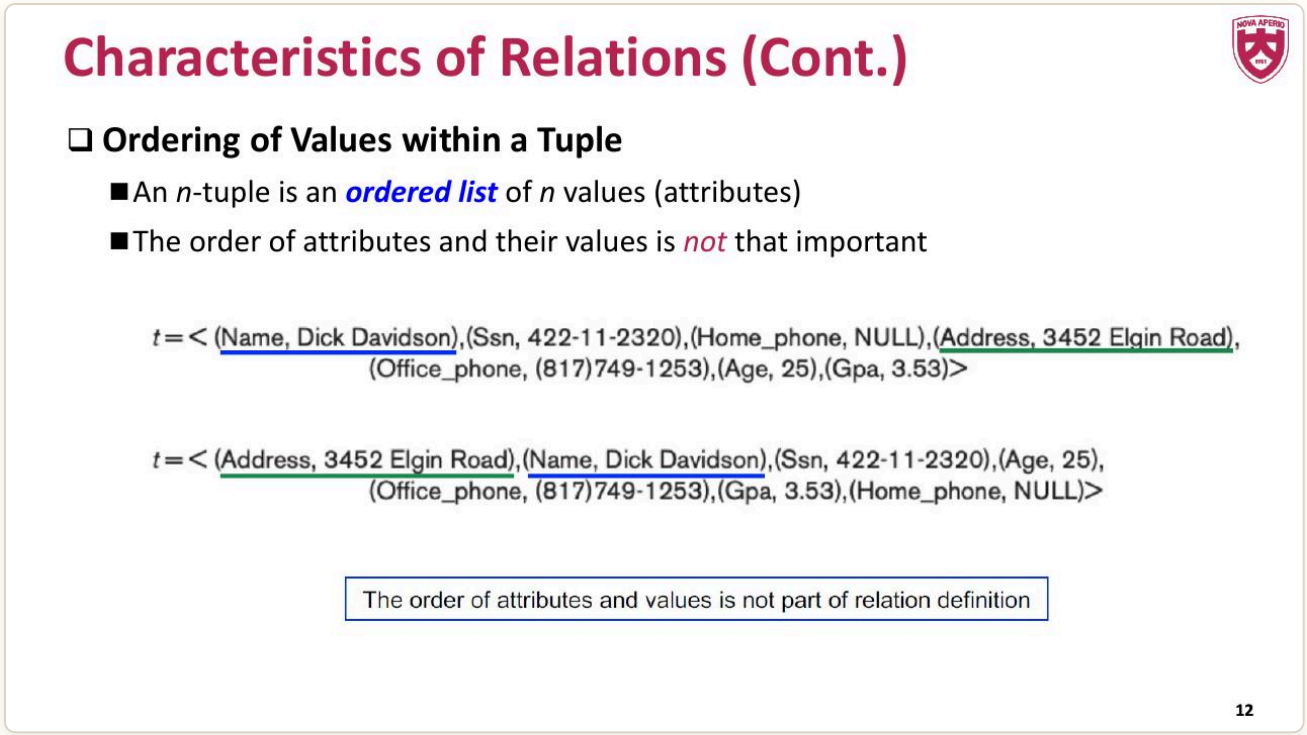
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릴레이션의 특성 ① 튜플의 순서

- 릴레이션은 튜플의 **집합(set)**으로 정의
  - Q. 집합에서 순서가 뜻이 있나?
  - 릴레이션의 튜플에는 **특정한 순서가 없다**
    - 그래서 tabular가 아니라 **relation**이라 부른다
- 그림: 순서 1과 순서 2는 **같은 릴레이션**
  - 표로 그려지고 디스크에 저장돼도 DBMS는 튜플 순서를 책임지지 않는다

필기 ✍️

★ 시험 튜플 순서 없음 (set) — O/X 단골



릴레이션의 특성 ② 튜플 안 값의 순서

- n-tuple은 n개 값의 **순서 있는 목록**
  - 속성과 값의 순서는 **그다지 중요하지 않다**
- 그림: (속성, 값) 쌍으로 쓰면 순서를 바꿔도 같은 튜플
  - 속성·값의 순서는 릴레이션 정의의 일부가 아니다

필기 ✍️

⚠ 주의 정의상 튜플은 '순서 있는 목록'이지만, (속성,값) 쌍으로 보면 순서는 중요하지 않다 — 둘 다 슬라이드 문장

Characteristics of Relations (Cont.)

❑ Values and NULLs in Tuples

- ■ Each value in a tuple is an ***atomic*** value
  - ❑ I.e., it is not divisible into components
- ■ *Composite and multivalued attributes are **NOT** allowed*
  - ❑ Multivalued attributes must be represented by *separate relations*
  - ❑ Composite attributes are represented only by *their simple component attributes*
- ■ NULL values are used to represent the values of attributes that may be unknown or may not apply to a tuple
  - ❑ Several meanings for NULL values: unknown, unavailable, undefined values, …

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

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릴레이션의 특성 ③ 값과 NULL

- 튜플의 각 값은 **원자 값(atomic)** — 더 나눌 수 없음
- **복합·다중값 속성은 허용되지 않는다**
  - 다중값 속성 → **별도 릴레이션**으로
  - 복합 속성 → **단순 구성 속성들**로만
- **NULL**: 모르거나 그 튜플에 해당하지 않는 속성 값
  - 뜻이 여럿: unknown, unavailable, undefined …

필기 ✍️

★ 시험 관계형 모델: composite·multivalued NOT allowed ⟶ 매핑 때 분해

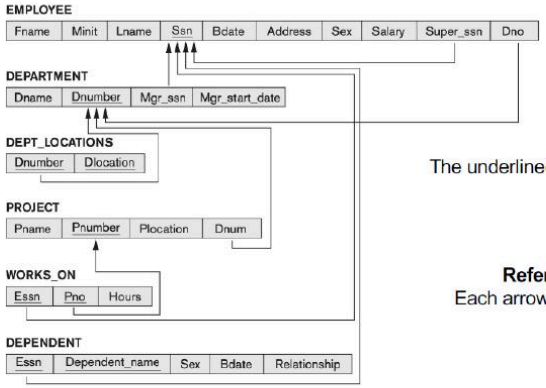
✏ 예.풀이 Office_Phone NULL = 사무실 전화가 없음(해당 없음) 또는 모름

Relational Database Schema

❑ What is a Relational Database Schema?

■ A set of **relation schemas** $S = \{R_1, \ldots, R_m\}$ and a set of ***integrity constraints*** *IC*

❑ E.g., COMPANY={EMPLOYEE, DEPARTMENT, DEPT_LOCATIONS, PROJECT, WORKS_ON, DEPEDENT}



Key Constraint:
The underlined attribute represents the primary key

Referential Integrity Constraint:
Each arrow represents the referential relation between two relations

Relational Database Schema for the COMPANY Database

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관계형 DB 스키마

- **Relational DB Schema**: 릴레이션 스키마의 집합 S = {R_1, ..., R_m} + **무결성 제약 집합 IC**
  - 예: COMPANY = {EMPLOYEE, DEPARTMENT, DEPT_LOCATIONS, PROJECT, WORKS_ON, DEPENDENT}
- 그림: 밑줄 = 기본키(키 제약), 화살표 = 참조 무결성 제약(두 릴레이션 사이 참조)

필기 ✍️

★ 시험 DB 스키마 = 릴레이션 스키마들 + IC (제약까지 포함!)

Relational Database State

❑ What is a Relational Database State?

■ A set of **relation states** $DB = \{r_1, \ldots, r_m\}$, r_i is a state of R_i

❑ r_i satisfies the integrity constraints specified in IC

❑ A DB state that does not obey all the integrity constraints is called **not valid**

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

One Possible DB State for the COMPANY Database

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관계형 DB 상태

- **Relational DB State**: 릴레이션 상태의 집합 DB = {r_1, ..., r_m}, r_i는 R_i의 상태
  - r_i는 IC의 무결성 제약을 만족
  - 모든 무결성 제약을 지키지 않는 DB 상태는 **not valid**
- 그림: COMPANY DB의 가능한 상태 하나

필기 ✍️

💡 이해 스키마 = 설계도(잘 안 바뀜), 상태 = 지금 데이터(자주 바뀜)

Outline

- ❑ **Relational Model Concepts**
  - ■ Domains, Attributes, Tuples, and Relations
  - ■ Characteristics of Relations, Relational Database and Schemas
- ❑ **Relational Model Constrains**
  - ■ Key Constraints
  - ■ Entity and Referential Integrity Constraints
- ❑ **Operations and Constraint Violations**
  - ■ Basic Operations: Insert, Update, Delete
  - ■ Constraint Violations
- ❑ **ER-to-Relational Mapping**
  - ■ Mapping of Entity Types
  - ■ Mapping of Multivalued Attributes
  - ■ Mapping of Relationship Types

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목차 (Outline)

- **관계형 모델 개념** — 도메인·속성·튜플·릴레이션 / 릴레이션의 특성, 관계형 DB와 스키마
- **관계형 모델 제약** — 키 제약 / 개체·참조 무결성
- **연산과 제약 위반** — Insert·Update·Delete / 제약 위반
- **ER → 관계 매핑** — 엔티티 타입 / 다중값 속성 / 관계 타입

필기 ✍️

💡 이해 두 번째 덩어리: 제약

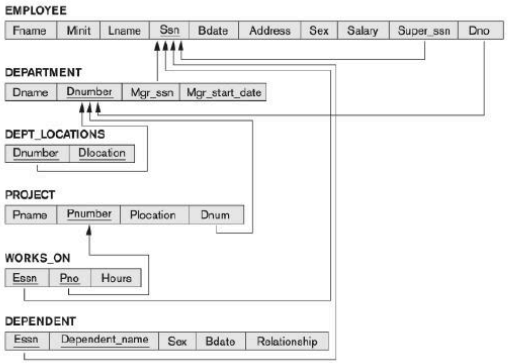
Relational Model Constraints

❑ **What are the Relational Model Constraints?**

- ■ There exit many relations, and their tuples are related in various way
- ■ Restrictions on the actual values should be specified
  - ❑ Derived from the rules (i.e., requirement analysis)
  - ❑ Every relation database state should follow all the constraints (i.e., valid state)

❑ **Three Crucial Constraints**

- ■ Key Constraints
- ■ Entity integrity Constraints
- ■ Referential integrity Constraints



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관계형 모델 제약

- 관계형 모델 제약이란?
  - 릴레이션이 많고, 튜플들이 여러 방식으로 관련됨
  - 실제 값에 대한 **제한**을 명시해야 함
    - 규칙(요구사항 분석)에서 나옴
    - 모든 DB 상태는 모든 제약을 따라야 함 (**valid state**)
- **핵심 제약 3가지**
  - **Key** constraints / **Entity integrity** constraints / **Referential integrity** constraints

필기 ✍️

★ 시험 제약 3개 이름 영어로 — Key · Entity integrity · Referential integrity

Key Constraints

❑ Why Key Constraints Need?

- In a relation, tuples must be ***distinct*** by the definition
- How can we identify each tuple uniquely?

❑ What is a Key Constraint?

- *Two distinct tuples cannot have the identical values for the attributes in SK*
  - For any two distinct tuples t_1 and t_2 in a relation state r: $t_1[SK] \neq t_2[SK]$
- *Superkey, SK,* of Relational Schema *R*
  - A ***subset of attributes*** **SK** in *R* that specifies the key constraint
  - Every relation has at least one default superkey

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

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키 제약 ① 슈퍼키

- 왜 필요한가?
  - 릴레이션의 튜플은 정의상 **서로 달라야** 함
  - 각 튜플을 어떻게 유일하게 식별하지?
- **Key constraint**: 서로 다른 두 튜플은 SK 속성들에 같은 값을 가질 수 없다
  - r의 서로 다른 t_1, t_2에 대해 $t_1[SK] \neq t_2[SK]$
- **Superkey SK**
  - 키 제약을 지정하는 R의 속성 부분집합
  - 모든 릴레이션은 **기본 슈퍼키를 최소 하나** 가진다

필기 ✍️

💡 이해 기본 슈퍼키 = 모든 속성 전체 (튜플이 서로 다르니까 항상 성립)

Key Constraints (Cont.)



❑ *Key* of Relational Schema *R*

- ■ *A superkey* of *R* with the following additional property
  - ❑ Removing any attribute ***A*** from ***K*** leaves a set of attributes ***K'*** that is not a superkey of R any more
- ■ It is a ***minimal*** superkey
  - ❑ We cannot remove any attributes, the uniqueness constraint holds
  - ❑ This *minimality* property is required for a key but optional for a superkey

Superkey: {Ssn, Home_phone}

Removing the attribute "Home_phone"

Key: {Ssn}

Superkey:
any set of attributes that include the key

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

The value of a key attribute can be used to identify uniquely each tuple

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키 제약 ② 키 = 최소 슈퍼키

- **Key**: 다음 성질을 더 가진 슈퍼키
  - K에서 어떤 속성 A를 빼도 남은 K'는 **더 이상 슈퍼키가 아님**
  - = **minimal superkey**
    - 어떤 속성도 뺄 수 없다 — 빼면 유일성이 깨짐
    - 이 **최소성은 키에는 필수, 슈퍼키에는 선택**

필기 ✍️

✏ 예.풀이 {Ssn, Name}은 슈퍼키지만 Name을 빼도 {Ssn}이 슈퍼키 ⟶ 키 아님. {Ssn}은 키

★ 시험 minimality: required for a key, optional for a superkey

Key Constraints (Cont.)



❑ *Primary and Candidate Keys* of Relational Schema *R*

- ■ A relation schema may have more than one key
  - ❑ Each of the keys is called a **candidate key** (i.e., minimal key)
- ■ It common to designate one of the candidate keys as the **primary key** of the relation
  - ❑ Used to *identify* tuples uniquely in the relation

There are two candidate keys in the CAR relation
License_number is designated as the primary key (by a database designer) for this relation

Primary key is underlined

CAR

License_number	Engine_serial_number	Make	Model	Year
Texas ABC-739	A69352	Ford	Mustang	02
Florida TVP-347	B43696	Oldsmobile	Cutlass	05
New York MPO-22	X83554	Oldsmobile	Delta	01
California 432-TFY	C43742	Mercedes	190-D	99
California RSK-629	Y82935	Toyota	Camry	04
Texas RSK-629	U028365	Jaguar	XJS	04

Usually, it is better to choose a primary key with a single attribute or a small number of attributes

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키 제약 ③ 후보키와 기본키

- 릴레이션 스키마는 **키를 여러 개** 가질 수 있다
  - 각 키를 **candidate key(후보키, = minimal key)**라 한다
  - 보통 후보키 중 하나를 **primary key(기본키)**로 지정
    - 릴레이션에서 튜플을 유일하게 식별하는 데 씀

필기 ✍️

★ 시험 포함 관계: 기본키 ⊂ 후보키 ⊂ 슈퍼키

✏ 예.풀이 CAR: Registration#,
Engine_serial# 둘 다 후보키 ⟶ 하나를 PK로

Entity Integrity Constraint

❑ **What is an Entity Integrity Constraint?**

- The constraint for the primary key (PK) of each relation schema *R*
- *No primary key value can be NULL*
  - For every tuple *t* in *r(R)*, *t[PK] ≠ NULL*

Office_phone cannot become the key

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
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Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

If two or more tuples have NULL for their primary keys,
we cannot distinguish them when we try to reference them from other relation

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개체 무결성 제약 (Entity Integrity)

- 각 릴레이션 스키마의 **기본키(PK)**에 대한 제약
  - **PK 값은 NULL일 수 없다**
    - r(R)의 모든 튜플 t에 대해 t[PK] ≠ NULL

필기 ✍️

★ 시험 Entity integrity = PK는 NULL 불가
(PK의 '일부'라도 NULL이면 위반)

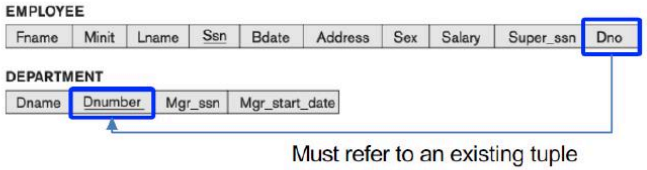
💡 이해 이름표가 비어 있으면 누군지 알 수 없다

Referential Integrity Constraint

❑ Why is a Referential Integrity Constraint Needed?

■ To maintain the consistency among tuples in the two relations

❑ A tuple in one relation referring to another relation must refer to an *existing tuple* in that relation



The attribute Dno of EMPLOYEE gives the department number for which each employee works

Thus, its value in every EMPLOYEE tuple must match the Dnumber value of some tuple in the DEPARTMENT relation

Q. What if the Dnumber is updated in the DEPARTMENT?

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참조 무결성 제약 ① 왜 필요한가

- 두 릴레이션의 튜플 사이 **일관성**을 지키려고
  - 다른 릴레이션을 참조하는 튜플은 그 릴레이션의 **존재하는 튜플**을 참조해야 한다

필기 ✍️

— 넘어가도 되는 쪽

Referential Integrity Constraint (Cont.)

❑ **What is a *Foreign Key*?**

■ A set of attributes FK in relation schema R_1 is a **foreign key** of R_1 that **references** R_2

❑ Specifying a referential integrity constraint between two relation schema R_1 and R_2

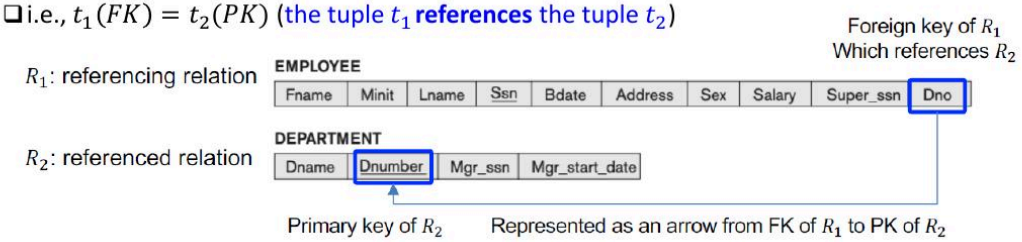
❑ **What is a Referential Integrity Constraint?**

■ The FK attributes of R_1 have the same domain(s) as the PK attributes of R_2

❑ The attributes FK are said to **reference** or **refer to** the relation R_2

■ A value of FK in a tuple t_1 either has a value of PK for some tuple t_2 or is NULL

❑ i.e., $t_1(FK) = t_2(PK)$ (the tuple t_1 **references** the tuple t_2)



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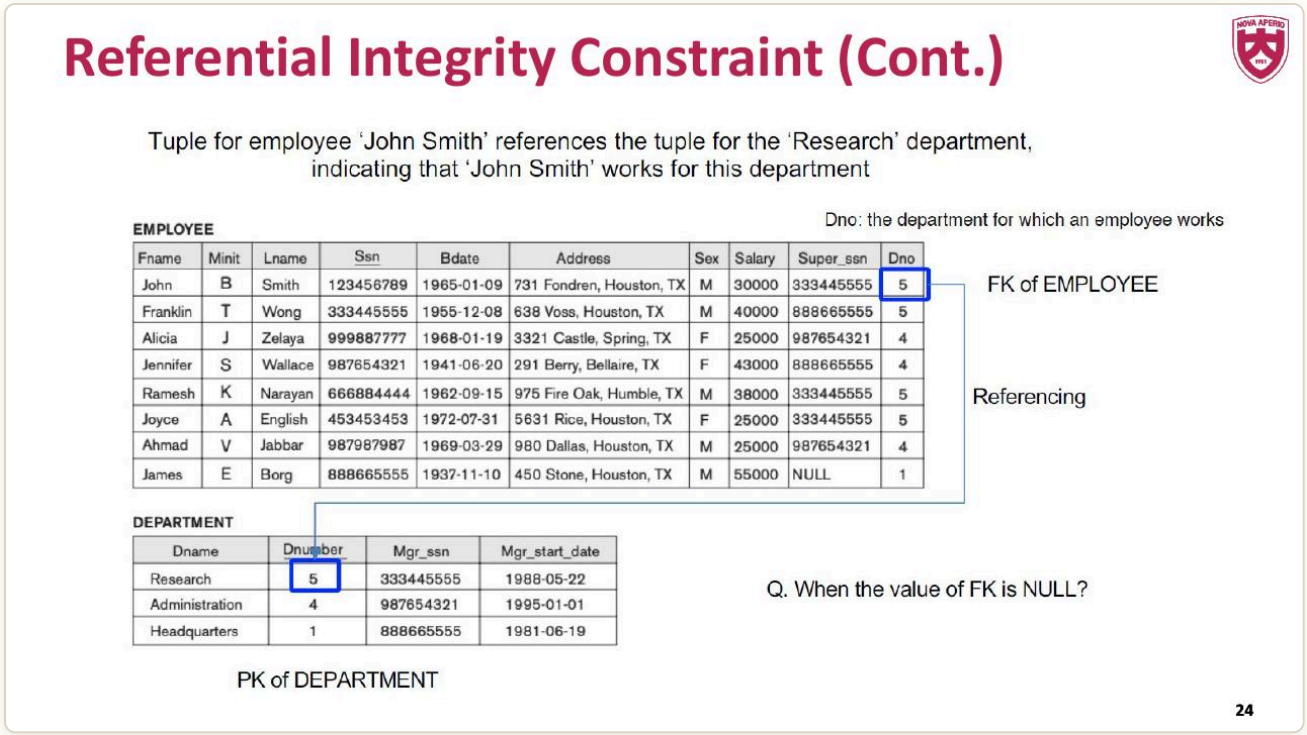
참조 무결성 제약 ② 외래키 (Foreign Key)

- **Foreign Key**: 릴레이션 R_1의 속성 집합 FK가 R_2를 참조
  - R_1과 R_2 사이의 참조 무결성 제약을 지정
- **Referential Integrity Constraint**
  - R_1의 FK 속성은 R_2의 PK 속성과 **같은 도메인**
    - FK가 R_2를 reference(refer to)한다고 함
  - 튜플 t_1의 FK 값은 어떤 튜플 t_2의 **PK 값**이거나 **NULL**
    - $t_1[FK] = t_2[PK]$ (t_1이 t_2를 참조)

필기 ✍️

★ 시험 FK 조건 2개: ① 같은 도메인 ② 참조하는 PK 값이 있거나 NULL

⚠ 주의 FK는 NULL이 될 수 있다! (PK와 다름)



참조 무결성 제약 (예)

- 예: EMPLOYEE 'John Smith'의 Dno = 5 (FK) → DEPARTMENT 'Research'의 Dnumber = 5 (PK)
  - Smith가 Research 부서에서 일한다는 뜻
- Q. FK가 NULL이면? → 아직 부서에 배정되지 않은 직원

필기 ✍️

💡 이해 화살표는 FK ⟶ 참조되는 PK 쪽으로 그린다

Exercises: Referential Integrity Constraints



❑ Exercise 1

■ Consider the following relations for a database that keeps track of business trips of salespersons in a sales office:

SALESPERSON(Ssn, Name, Start_year, Dept_no)
TRIP(Ssn, From_city, To_city, Departure_date, Return_date, Trip_id)
EXPENSE(Trip_id, Account#, Amount)



■ A trip can be charged to one or more accounts

■ Specify all foreign keys for this schema

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연습 1: 외래키 찾기

- 영업 사원의 출장을 기록하는 DB
  - SALESPERSON(**Ssn**, Name, Start_year, Dept_no)
  - TRIP(Ssn, From_city, To_city, Departure_date, Return_date, **Trip_id**)
  - EXPENSE(**Trip_id, Account#**, Amount)
  - 출장 하나는 계정 하나 이상에 청구될 수 있다
  - 모든 외래키를 적어라

필기 ✍️

✏ 예.풀이 답(p.58): TRIP.Ssn ⟶ SALESPERSON · EXPENSE.Trip_id ⟶ TRIP

⚠ 주의 Dept_no는 참조할 DEPARTMENT 표가 스키마에 없으므로 FK 아님

Exercises: Referential Integrity Constraints

❑ **Exercise 2**

■ Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

STUDENT(Ssn, Name, Major, Bdate)
COURSE(Course#, Cname, Dept)
ENROLL(Ssn, Course#, Quarter, Grade)
BOOK_ADOPTION(Course#, Quarter, Book_isbn)
TEXT(Book_isbn, Book_title, Publisher, Author)



■ Specify all foreign keys for this schema

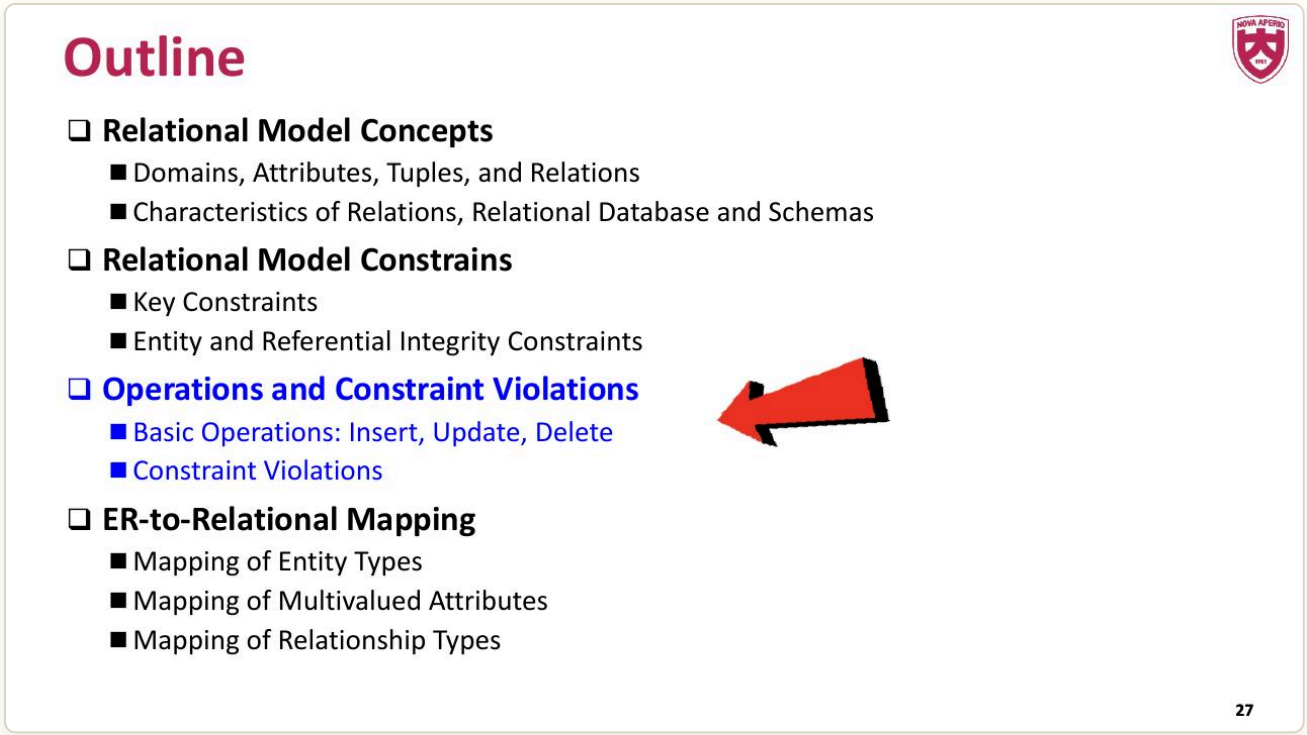
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연습 2: 외래키 찾기

- 수강과 교재를 기록하는 DB
  - STUDENT(**Ssn**, Name, Major, Bdate)
  - COURSE(**Course#**, Cname, Dept)
  - ENROLL(**Ssn, Course#, Quarter**, Grade)
  - BOOK_ADOPTION(**Course#, Quarter**, Book_isbn)
  - TEXT(**Book_isbn**, Book_title, Publisher, Author)
  - 모든 외래키를 적어라

필기 ✍️

✏ 예.풀이 답(p.59): ENROLL.Ssn ⟶ STUDENT
· ENROLL.course# ⟶ COURSE ·
BOOK_ADOPTION.course# ⟶ COURSE ·
BOOK_ADOPTION.Book_isbn ⟶ TEXT



목차 (Outline)

- **관계형 모델 개념** — 도메인·속성·튜플·릴레이션 / 릴레이션의 특성, 관계형 DB와 스키마
- **관계형 모델 제약** — 키 제약 / 개체·참조 무결성
- **연산과 제약 위반** — Insert·Update·Delete / 제약 위반
- **ER → 관계 매핑** — 엔티티 타입 / 다중값 속성 / 관계 타입

필기 ✍️

💡 이해 세 번째 덩어리: 연산과 제약 위반

Operations of Relational Model

❑ **Two Types of Operations**

- ■ **Retrieval**: to retrieve information from the database
- ■ **Update**: to change the states of relations in the database

❑ **Basic Update Operations**

- ■ **Insert**: insert one or more new tuples into a relation
- ■ **Delete**: remove tuples from a relation
- ■ **Update** (or **Modify**): change attribute values in existing tuples



Whenever these operations are applied, some constraints may be violated

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관계형 모델의 연산

- 연산 두 종류
  - **Retrieval** — DB에서 정보 꺼내기
  - **Update** — DB 릴레이션 상태 바꾸기
- 기본 갱신 연산
  - **Insert** — 릴레이션에 새 튜플(들) 추가
  - **Delete** — 튜플 제거
  - **Update(Modify)** — 기존 튜플의 속성 값 변경
- 이 연산을 할 때마다 **제약이 깨질 수 있다**

필기 ✍️

— 넘어가도 되는 쪽

Basic Update Operation: Insert

❑ Insert Operation

■ Add a single tuple (row) to a relation (table)

❑ Need to specify the relation name and a list of values for the tuple

Operation:
Insert <'Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, NULL, 4> into **EMPLOYEE.**

Insert a tuple with all attribute values
The values should be listed in the same order

Q. What kinds of constraints can be violated by Insert operations?

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

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기본 갱신 연산: Insert

- 릴레이션(표)에 튜플(행) 하나 추가
  - 릴레이션 이름과 튜플 값 목록을 지정
- 예: Insert <'Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, NULL, 4> into EMPLOYEE

필기 ✍️

— 넘어가도 되는 쪽

Basic Update Operation: Insert (Cont.)

❑ Constraint Violations by the Insert Operation

- ■ **Key constraint**: a key value in the new tuple already exist in another tuple
- ■ **Entity constraint**: any part of the PK of the new tuple is NULL

Operation:
Insert <'Alicia', 'J', 'Zelaya', '999887777', '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, '987654321', 4> into EMPLOYEE.

→ This operation violates the key constraint

Operation:
Insert <'Cecilia', 'F', 'Kolonsky', NULL, '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, NULL, 4> into EMPLOYEE.

→ This operation violates the entity integrity constraint

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

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Insert의 제약 위반 ①

- **Key constraint**: 새 튜플의 키 값이 이미 다른 튜플에 있음
  - 예: Ssn '999887777'(이미 Zelaya) 삽입 → 키 제약 위반
- **Entity constraint**: 새 튜플 PK의 **어느 부분이라도 NULL**
  - 예: Ssn NULL 삽입 → 개체 무결성 위반

필기 ✍️
— 넘어가도 되는 쪽

Basic Update Operation: Insert (Cont.)

❑ Constraint Violations by the Insert Operation

■ **Referential integrity constraint**: referenced tuples do not exist

Operation:
Insert <'Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windswept, Katy, TX', F, 28000, '987654321', 7> into EMPLOYEE.

→ This operation violates the referential integrity constraint

Operation:
Insert <'Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, NULL, 4> into EMPLOYEE.

→ This operation satisfies all constraints

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

There is no tuple with Dnumber=7

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Insert의 제약 위반 ②

- **Referential integrity**: 참조할 튜플이 없음
  - 예: Dno = 7 삽입 → Dnumber 7 부서 없음 → 참조 무결성 위반
  - 예: Dno = 4로 올바르게 삽입 → 모든 제약 만족

필기 ✍️

★ 시험 Insert는 세 제약 모두 깰 수 있다 (키·개체·참조)

Basic Update Operation: Delete

❑ Delete Operation

■ Remove a tuple from a relation

❑ Need to specify the relation name and the values of PK attributes

Operation:
Delete the WORKS_ON tuple with Essn = '999887777' and Pno = 10.

Q. What kinds of constraints can be violated by Delete operations?

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

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기본 갱신 연산: Delete

- 릴레이션에서 튜플 제거
  - 릴레이션 이름과 **PK 값**을 지정

필기 ✍️

— 넘어가도 되는 쪽

Basic Update Operation: Delete (Cont.)

❑ Constraint Violations by the Delete Operation

■ **Referential integrity constraint**: the deleted tuple is referenced from other tuples

Operation:
Delete the EMPLOYEE tuple with Ssn = '999887777'

Two tuples in WORKS_ON refers to this tuple
→ Violate the referential integrity constraint

Operation:
Delete the EMPLOYEE tuple with Ssn = '333445555'

Four tuples in WORKS_ON refers to this tuple
Two tuples in EMPLOYEE refers to this tuple
→ Violate the referential integrity constraint

Operation:
Delete the WORKS_ON tuple with Essn = '999887777' and Pno = 10.

No tuples refers to this tuple
→ This operation satisfies the referential integrity constraint

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

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Delete의 제약 위반

- **Referential integrity**: 지운 튜플을 **다른 튜플이 참조**하고 있음
  - 예: EMPLOYEE Ssn='999887777' 삭제 → WORKS_ON 두 튜플이 참조 → 위반
  - 예: EMPLOYEE Ssn='333445555' 삭제 → WORKS_ON·EMPLOYEE 여러 튜플이 참조 → 위반
  - 예: WORKS_ON (999887777, 10) 삭제 → 참조하는 곳 없음 → 만족

필기 ✍️

★ 시험 Delete는 참조 무결성만 깬다

💡 이해 대처(4강): RESTRICT · CASCADE · SET NULL · SET DEFAULT

Basic Update Operation: Update

❑ Update (Modify) Operation

- ■ Modify the values of one or more attributes
  - ❑ Need to specify the attributes being updated and their values

Operation:
Update the salary of the EMPLOYEE tuple with Ssn = '999887777' to 28000.

Operation:
Update the Dno of the EMPLOYEE tuple with Ssn = '999887777' to 1.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Q. What kinds of constraints can be violated by Update operations?

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기본 갱신 연산: Update

- 속성 하나 이상의 값을 바꿈
  - 바꿀 속성과 새 값을 지정

필기 ✍️
— 넘어가도 되는 쪽

Basic Update Operation: Update (Cont.)



❑ **UPDATE examples that may violate some constraints**

Operation:
Update the Dno of the EMPLOYEE tuple with Ssn = '999887777' to 7.

→ This operation violates the referential integrity constraint

Operation:
Update the Ssn of the EMPLOYEE tuple with Ssn = '999887777' to '987654321'

→ This operation violates both the key and the referential integrity constraints

Only updating the attributes of PK or FK can cause the problems

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

Update의 제약 위반 (예)

- 예: Ssn='999887777'의 Dno를 7로 → 참조 무결성 위반
- 예: Ssn '999887777'을 '987654321'로 → 키 + 참조 무결성 위반
- 예: Salary 변경 → 만족

필기 ✍️

★ 시험 Update는 키·참조를 깰 수 있다. PK나 FK를 바꿀 때만 문제

⚠ 주의 PK를 NULL로 바꾸면 개체 무결성도 깨짐 — 하지만 슬라이드 요약은 '키·참조'

Exercises: Constraint Violation



❑ Exercise 3

- ■ Suppose each of the following basic operations is applied directly to the database state shown in the next slide
- ■ Describe all integrity constraints violated by each operation

a. Insert <'Robert', 'F', 'Scott', '943775543', '1972-06-21', '2365 Newcastle Rd, Bellaire, TX', M, 58000, '888665555', 1> into EMPLOYEE.
b. Insert <'ProductA', 4, 'Bellaire', 2> into PROJECT.
c. Insert <'Production', 4, '943775543', '2007-10-01'> into DEPARTMENT.
d. Insert <'677678989', NULL, '40.0'> into WORKS_ON.
e. Insert <'453453453', 'John', 'M', '1990-12-12', 'spouse'> into DEPENDENT.
f. Delete the WORKS_ON tuples with Essn = '333445555'.
g. Delete the EMPLOYEE tuple with Ssn = '987654321'.
h. Delete the PROJECT tuple with Pname = 'ProductX'.
i. Modify the Mgr_ssn and Mgr_start_date of the DEPARTMENT tuple with Dnumber = 5 to '123456789' and '2007-10-01', respectively.
j. Modify the Super_ssn attribute of the EMPLOYEE tuple with Ssn = '999887777' to '943775543'.
k. Modify the Hours attribute of the WORKS_ON tuple with Essn = '999887777' and Pno = 10 to '5.0'.

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연습 3: 제약 위반 ①

- 아래 연산을 다음 쪽 DB 상태에 바로 적용한다고 하자
  - 각 연산이 어기는 **무결성 제약을 모두** 적어라
  - (a)~(k) 연산 목록

필기 ✍️

★ 시험 답은 P.60~70에 한 쪽씩 — 이 단원 최고 빈출 유형

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

- Suppose each of the following basic operations is applied directly to the database state shown in the next slide
- Describe all integrity constraints violated by each operation

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3: 제약 위반 ②

- 연습 3의 COMPANY DB 상태 (p.7과 같은 표들)

필기 ✍️

💡 이해 풀 때 순서: ① PK 중복? ② PK NULL? ③ FK가 가리키는 행 있나? ④ 삭제면 나를 가리키는 행 있나?

Outline

- ❑ **Relational Model Concepts**
  - ■ Domains, Attributes, Tuples, and Relations
  - ■ Characteristics of Relations, Relational Database and Schemas
- ❑ **Relational Model Constrains**
  - ■ Key Constraints
  - ■ Entity and Referential Integrity Constraints
- ❑ **Operations and Constraint Violations**
  - ■ Basic Operations: Insert, Update, Delete
  - ■ Constraint Violations
- ❑ **ER-to-Relational Mapping**
  - ■ Mapping of Entity Types
  - ■ Mapping of Multivalued Attributes
  - ■ Mapping of Relationship Types



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목차 (Outline)

- **관계형 모델 개념** — 도메인·속성·튜플·릴레이션 / 릴레이션의 특성, 관계형 DB와 스키마
- **관계형 모델 제약** — 키 제약 / 개체·참조 무결성
- **연산과 제약 위반** — Insert·Update·Delete / 제약 위반
- **ER → 관계 매핑** — 엔티티 타입 / 다중값 속성 / 관계 타입

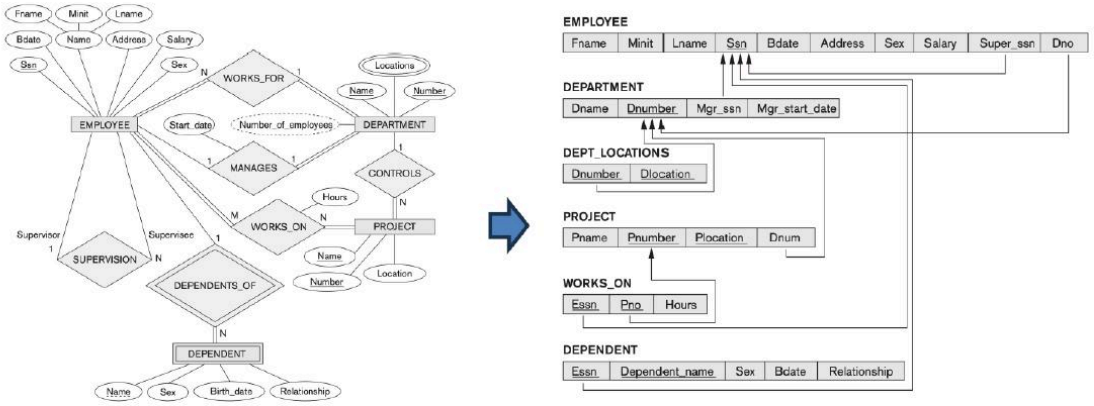
필기 ✍️

💡 이해 네 번째 덩어리: ER → 관계 매핑 (시험 서술형 단골)

Relational Database Design

❑ ER-to-Relational Mapping

■ Let's design a relational database schema based on a conceptual schema design



ER-to-Relational Mapping

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관계형 DB 설계

- **ER-to-Relational Mapping**
  - 개념 스키마 설계(ER)를 바탕으로 관계형 DB 스키마를 설계하자
- 그림: COMPANY ER 다이어그램 → 관계형 스키마

필기 ✍️

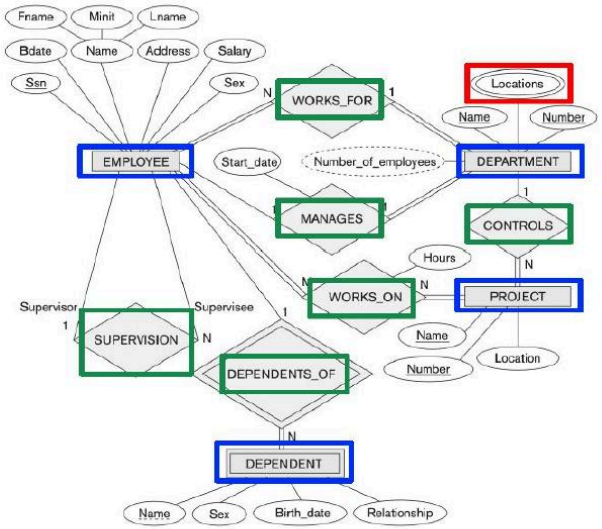
— 넘어가도 되는 쪽

Process of ER-to-Relational Mapping

❑ **Mapping of Entity Types**
- Regular and weak entity types

❑ **Mapping of Multivalued Attributes**
- Multivalued attributes are not allowed

❑ **Mapping of Relationship Types**
- Binary relationship types: 1:1, 1:N, M:N
- N-ary relationship types (e.g., ternary)



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ER → 관계 매핑 과정

- **엔티티 타입 매핑** — 일반·약한 엔티티 타입
- **다중값 속성 매핑** — 관계형에선 다중값이 허용되지 않음
- **관계 타입 매핑** — 이항 1:1, 1:N, M:N / N항(3항 등)

필기 ✍️

★ 시험 순서: 일반 엔티티 → 다중값 → 약한 엔티티 → 1:1 → 1:N → M:N → N항

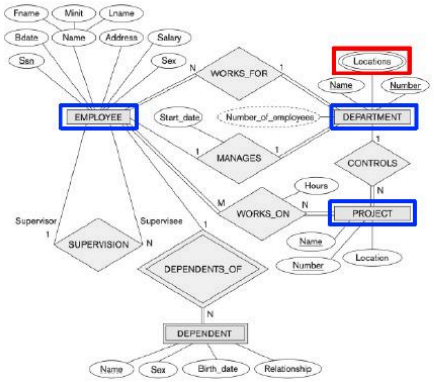
Process 1: Regular Entity Types

❑ Mapping a Regular Entity Type

■ (1) Create a relation *R* that includes all the simple attributes of the entity type *E*

❑ For a composite attribute, include only the simple component attributes of the composite attribute

■ (2) Designate one of the key attributes of *E* as the primary key for *R*



EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary

DEPARTMENT ?

Dname	Dnumber	Locations

How do we map a multivalued attribute?

PROJECT

Pname	Pnumber	Plocation

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과정 1: 일반 엔티티 타입

- (1) 엔티티 타입 E의 **모든 단순 속성**을 담은 릴레이션 R 생성
  - 복합 속성은 **단순 구성 속성만** 넣는다
- (2) E의 키 속성 중 하나를 R의 **기본키**로
- 예: EMPLOYEE(Fname, Minit, Lname, **Ssn**, Bdate, Address, Sex, Salary)
  - DEPARTMENT(Dname, **Dnumber**) — Locations(다중값)는? / PROJECT(Pname, **Pnumber**, Plocation)

필기 ✍️

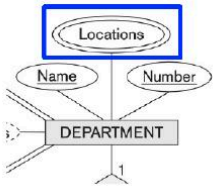
✏ 예.풀이 Name(Fname, Minit, Lname)은 Name 대신 세 칸으로

Process 1-2: Multivalued Attributes

❑ **Mapping a Multivalued Attribute**

- ■(1) Create a *separate* relation *R* for each multivalued attribute A
- ■(2) Includes the attribute *A* and the primary key of the original relation
  - ❑Designate the original relation's primary key as a **foreign key** in R
  - ❑The key attribute value of the entity (FK) is repeated

❑ (3) Set the combination of *A* and *K* as the primary key



DEPARTMENT

Dname	Dnumber

DEPT_LOCATIONS

Dnumber	Dlocation
5	Seoul
5	Incheon
5	Suwon
3	Seoul
3	Daejeon

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과정 1-2: 다중값 속성

- (1) 다중값 속성 A마다 **별도 릴레이션 R** 생성
- (2) 속성 A와 **원래 릴레이션의 기본키 K**를 넣는다
  - K를 R의 **외래키**로 — 엔티티의 키 값이 반복됨
- (3) **A와 K의 조합**을 기본키로
- 예: DEPT_LOCATIONS(**Dnumber, Dlocation**) — (5, Seoul), (5, Incheon), (3, Seoul) …

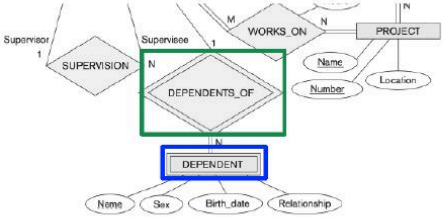
필기 ✍️

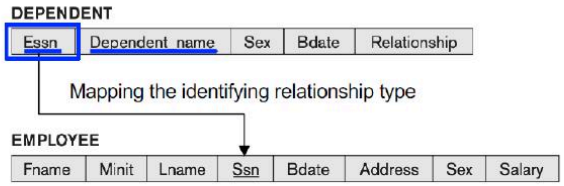
★ 시험 다중값 → 새 표(원래 PK + 그 속성), PK = 둘의 조합

Process 2: Weak Entity Types

❑ Mapping a Weak Entity Type

- ■(1) Create a relation *R* including all the simple attributes of the weak entity type *W*
  - ❑For a composite attribute, include only the simple component attributes of the composite attribute
- ■(2) Include the primary key of the owner entity type as a ***foreign key***
  - ❑This takes care of mapping the identifying relationship type of *W*
- ■(3) Set the combination of the owner's PK and the partial key as the primary key





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과정 2: 약한 엔티티 타입

- (1) 약한 엔티티 타입 W의 모든 단순 속성을 담은 릴레이션 R (복합은 구성 속성만)
- (2) **owner 엔티티 타입의 기본키를 외래키로** 포함
  - 이것으로 W의 **식별 관계**까지 매핑됨
- (3) **owner의 PK + 부분키** 조합을 기본키로
- 예: DEPENDENT(**Essn, Dependent_name**, Sex, Bdate, Relationship), Essn → EMPLOYEE.Ssn

필기 ✍️

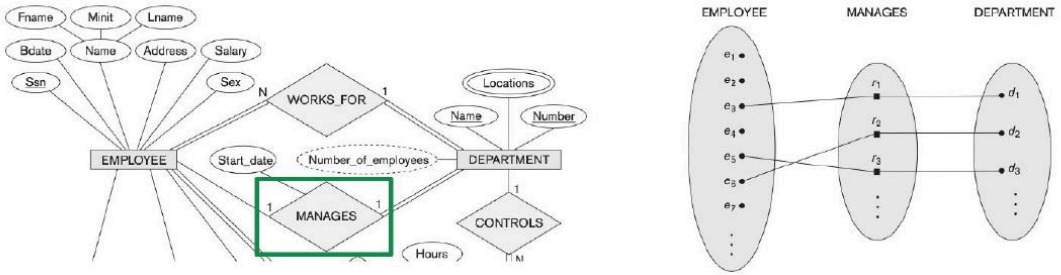
★ 시험 약한 엔티티 PK = owner PK + partial key

💡 이해 식별 관계(DEPENDENTS_OF)는 따로 표를 만들지 않는다

Process 3-1: Binary Relationship Types (1:1)

❑ **Three Possible Approaches**

- ■ ***Foreign key approach*** (*most frequently used and useful*)
- ■ Merged relation approach
- ■ Cross-reference (relationship relation) approach



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과정 3-1: 이항 1:1 관계

- 세 가지 방법
  - **Foreign key approach** — 가장 자주 쓰고 유용
  - **Merged relation approach** — 두 엔티티를 한 표로 합침
  - **Cross-reference (relationship relation) approach** — 관계용 표를 따로

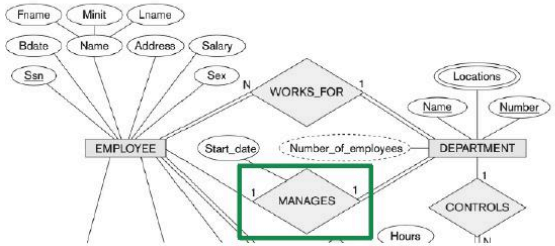
필기 ✍️

★ 시험 1:1 방법 3개 이름 외우기

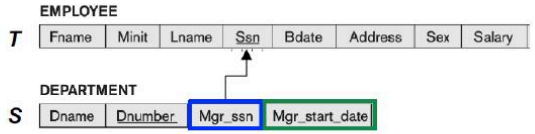
Process 3-1: Binary Relationship Types (1:1)

❑ Foreign Key Approach

- ■(1) Identify the participating entity types *S* and *T* of the relationship type R
- ■(2) Choose one of the relations and include the PK of the other one as an FK
  - ❑It is better to choose an entity type with *total participation in* the role of *S* (Why?)
- ■(3) Include simple attributes of the relationship type *R* as attributes of *S*



Relationship type attributes can be migrated into one of any participating entities



The entity type DEPARTMENT is chosen as the role of *S*

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과정 3-1: 1:1 — 외래키 방법 ①

- (1) 관계 R의 참여 엔티티 타입 S와 T를 확인
- (2) 한쪽을 골라 **다른 쪽의 PK를 FK로** 넣는다
  - **전체 참여(total)인 엔티티를 S로** 고르는 게 낫다 (왜?)
- (3) 관계 R의 단순 속성을 S의 속성으로

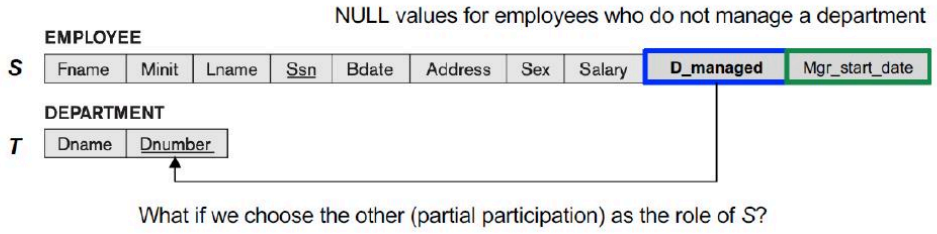
필기 ✍️

★ 시험 왜 total 쪽? ⟶ partial 쪽에 넣으면 FK가 NULL인 행이 많아짐

Process 3-1: Binary Relationship Types (1:1)

❑ Foreign Key Approach

- ■(1) Identify the participating entity types *S* and *T* of the relationship type R
- ■(2) Choose one of the relations and include the PK of the other one as an FK
  - ❑It is better to choose an entity type with *total participation in* the role of *S* (Why?)
- ■(3) Include simple attributes of the relationship type *R* as attributes of *S*



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과정 3-1: 1:1 — 외래키 방법 ② (예)

- 예: MANAGES — DEPARTMENT가 total → S = DEPARTMENT
  - DEPARTMENT(Dname, **Dnumber**, **Mgr_ssn**, **Mgr_start_date**), Mgr_ssn → EMPLOYEE.Ssn
  - 관계 속성 Start_date도 DEPARTMENT에

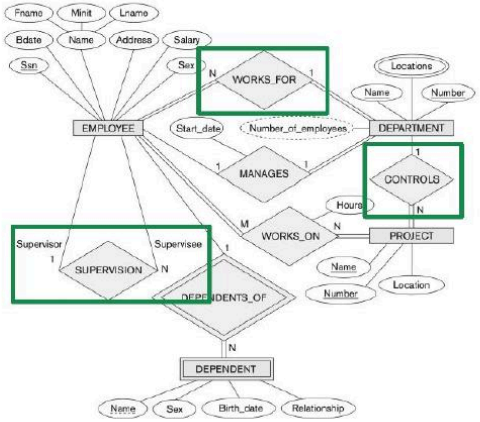
필기 ✍️

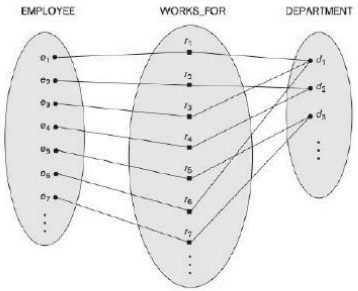
✏ 예.풀이 EMPLOYEE 쪽에 넣으면 관리자가 아닌 대부분 직원의 칸이 NULL

Process 3-2: Binary Relationship Types (1:N)

❑ **Two Possible Approaches**

- ***Foreign key approach*** (preferred as it reduces the number of relations)
- Cross-reference (relationship relation) approach





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과정 3-2: 이항 1:N 관계

- 두 가지 방법
  - **Foreign key approach** — 릴레이션 수를 줄여서 선호
  - **Cross-reference approach**

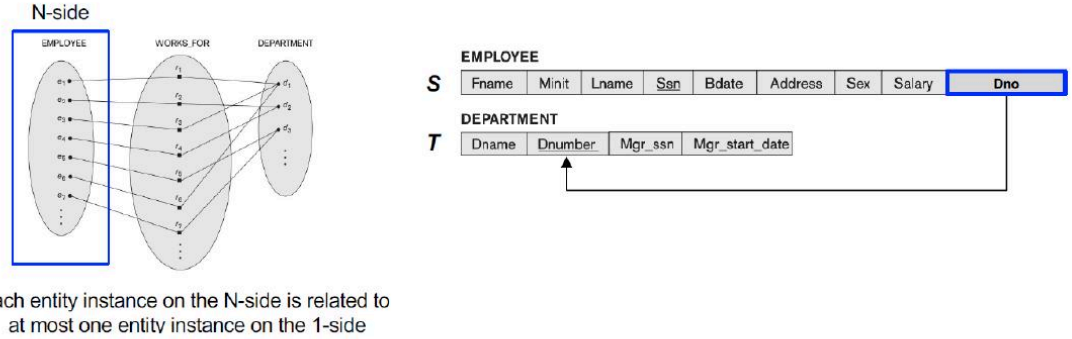
필기 ✍️

— 넘어가도 되는 쪽

Process 3-2: Binary Relationship Types (1:N)

❑ Foreign Key Approach

- (1) Identify the relation *S* representing the ***N-side*** entity type of the relationship type
- (2) Include the primary key of the other relation *T* as a foreign key in *S*
- (3) Include simple attributes of the relationship type as attributes of *S*



Each entity instance on the N-side is related to at most one entity instance on the 1-side

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과정 3-2: 1:N — 외래키 방법 ①

- (1) **N쪽** 엔티티 타입을 나타내는 릴레이션 S를 찾는다
- (2) 다른 릴레이션 T의 **PK를 S의 FK로**
- (3) 관계의 단순 속성을 S의 속성으로

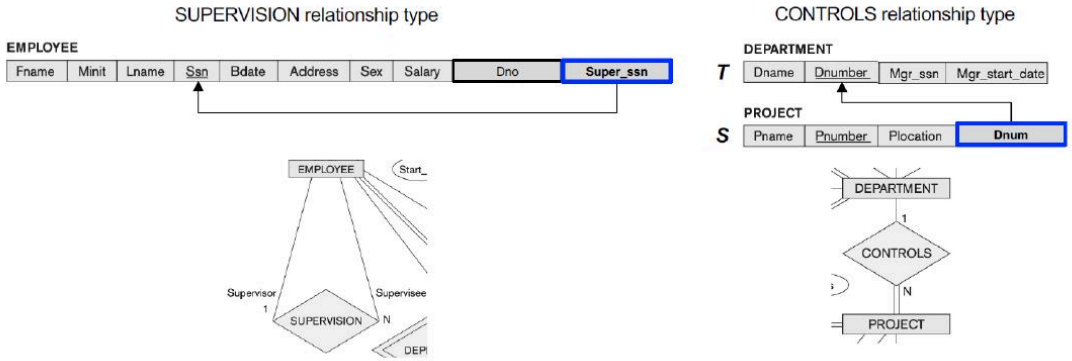
필기 ✍️

★ 시험 1:N ⟶ FK는 N쪽에! (1쪽에 넣으면 한 하나에 값이 여러 개)

Process 3-2: Binary Relationship Types (1:N)

❑ Foreign Key Approach

- (1) Identify the relation *S* representing the ***N-side*** entity type of the relationship type
- (2) Include the primary key of the other relation *T* as a foreign key in *S*
- (3) Include simple attributes of the relationship type as attributes of *S*



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과정 3-2: 1:N — 외래키 방법 ② (예)

- 예: WORKS_FOR — EMPLOYEE(..., **Dno**), Dno → DEPARTMENT.Dnumber
  - SUPERVISION — EMPLOYEE(..., **Super_ssn**), CONTROLS — PROJECT(..., **Dnum**)
  - N쪽의 각 인스턴스는 1쪽의 **최대 하나**와 관련 → FK 칸 하나로 충분

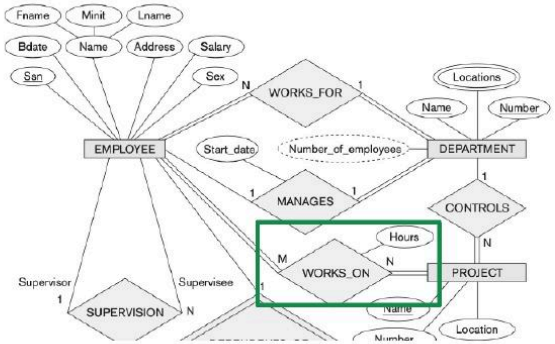
필기 ✍️

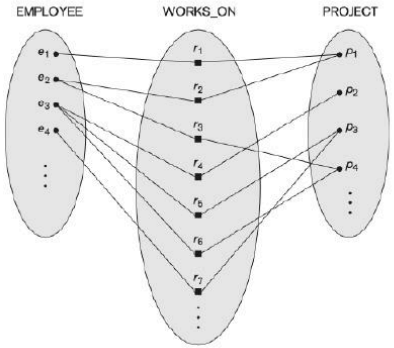
✏ 예.풀이 Super_ssn은 같은 EMPLOYEE 표를 가리키는 FK (순환 관계)

Process 3-3: Binary Relationship Types (M:N)

❑ **Only One Possible Approach**

■ **Cross-reference (relationship relation) approach**





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과정 3-3: 이항 M:N 관계

- 방법은 **하나뿐**: Cross-reference (relationship relation) approach

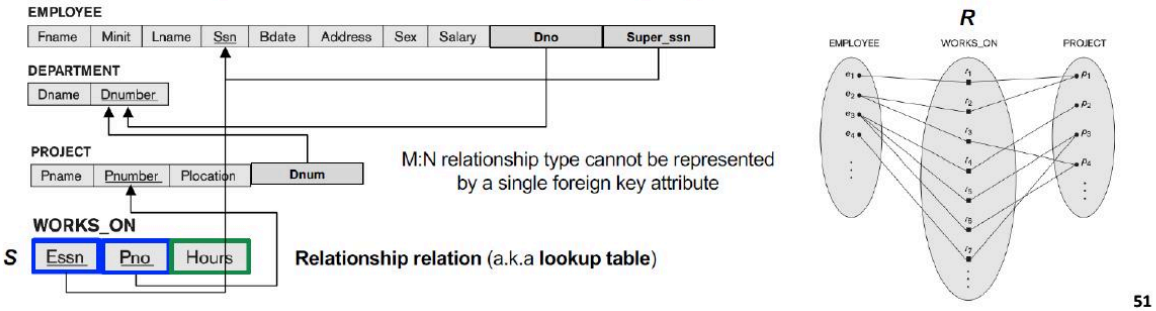
필기 ✍️

★ 시험 M:N은 FK 한 칸으로 못 나타낸다 ⟶ 반드시 새 표

Process 3-3: Binary Relationship Types (M:N)

❑ **Cross-Reference Approach**

- (1) Create a separate relation *S* to represent the relationship type *R*
- (2) Include *the primary keys of the relations representing the participating entity types as foreign keys* in *S*
- (3) Set the combination of all the foreign keys as the primary key
- (4) Include simple attributes of M:N relationship type as attributes of *S*



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과정 3-3: M:N — 교차 참조 방법

- (1) 관계 타입 R을 나타낼 **별도 릴레이션 S** 생성
- (2) 참여 엔티티 타입 릴레이션들의 **PK를 S의 FK로**
- (3) **모든 FK의 조합을 PK로**
- (4) M:N 관계의 단순 속성을 S에
- 예: WORKS_ON(**Essn, Pno**, Hours) — relationship relation (lookup table)

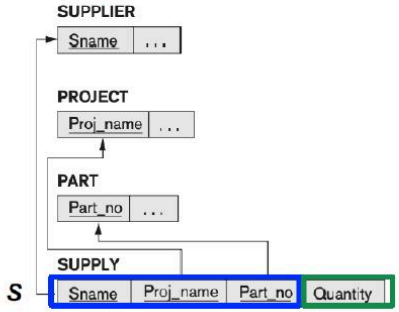
필기 ✍️

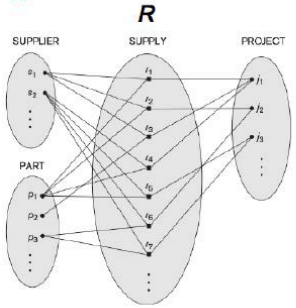
★ 시험 WORKS_ON PK = (Essn, Pno), Hours는 관계 속성으로 여기에

Process 4: N-ary Relationship Types

❑ Cross-Reference Approach

- ■(1) Create a separate relation *S* to represent the relationship type *R*
- ■(2) Include *the primary keys of the relations representing the participating entity types as foreign keys* in *S*
- ■(3) Set the combination of all the foreign keys as the primary key
- ■(4) Include simple attributes of M:N relationship type as attributes of *S*





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과정 4: N항 관계

- Cross-reference approach
  - (1) 별도 릴레이션 S
  - (2) 참여 엔티티들의 PK를 FK로
  - (3) 모든 FK 조합을 PK로
  - (4) 관계의 단순 속성을 S에
- 예: SUPPLY(**Sname, Proj_name, Part_no**, Quantity) — SUPPLIER, PROJECT, PART 참조

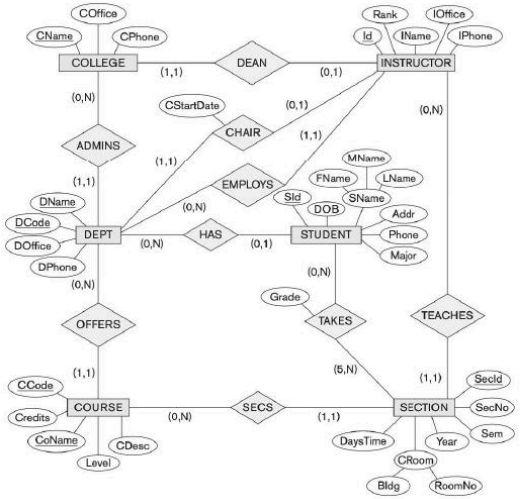
필기 ✍️

💡 이해 M:N과 똑같은 방법, 키만 n개

Exercises: ER-to-Relational Mapping

❑ Exercise 4

■ Map the ER schema for UNIVERSITY database into a relational database schema



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연습 4: ER → 관계 매핑

- UNIVERSITY DB의 ER 스키마를 관계형 스키마로 매핑하라
  - 엔티티: COLLEGE, INSTRUCTOR, DEPT, COURSE, SECTION, STUDENT
  - 관계: DEAN, CHAIR, ADMINS, EMPLOYS, HAS, OFFERS, SECS, TEACHES, TAKES

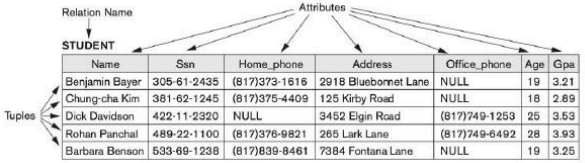
필기 ✍️

✏ 예.풀이 풀이는 p.71~74 (Step 1 엔티티 ⟶ 2 1:1 ⟶ 3 1:N ⟶ 4 M:N)

Summary: Relational Data Model

❑ Relational Model Concepts

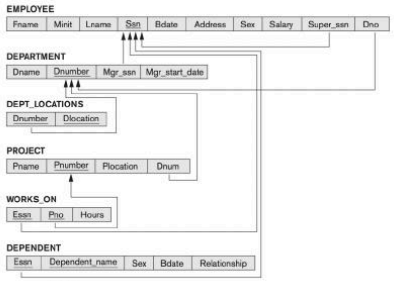
- ■ Relations, tuples, attributes, domains
- ■ Characteristics of Relations
- ■ Relational database schema and states



Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

❑ Relational Model Constraints

- ■ Key Constraints
  - ❑ *Two distinct tuples cannot have the identical values for the key attributes*
- ■ Entity Integrity Constraints
  - ❑ *No primary key value can be NULL*
- ■ Referential Integrity Constraints
  - ❑ The FK attributes of R_1 have the same domain(s) as the PK attributes of R_1
  - ❑ A value of FK in a tuple either has a value of PK for some tuple or is NULL



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요약: 관계형 데이터 모델

- 개념 — relation, tuple, attribute, domain / 릴레이션의 특성 / 관계형 DB 스키마와 상태
- 제약
  - **Key** — 서로 다른 두 튜플이 키 값이 같을 수 없음
  - **Entity integrity** — PK 값은 NULL 불가
  - **Referential integrity** — FK는 PK와 같은 도메인, 어떤 튜플의 PK 값이거나 NULL

필기 ✍️

💡 이해 슬라이드의 'PK attributes of R_1'은 R_2의 오타

Summary: Relational Data Model (Cont.)

❑ **Basic Update Operations**

- ■ **Insert**: insert one or more new tuples into a relation
- ■ **Delete**: remove tuples from a relation
- ■ **Update** (or **Modify**): change attribute values in existing tuples

❑ **Constraint Violations**

- ■ **Insert**: may violate any of constraints (key, entity integrity, and referential integrity)
- ■ **Delete**: may violate the referential integrity constraint
- ■ **Update**: may violate the key and referential integrity constraint
  - ❑ Only updating the attributes of PK or FK can cause the problems





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요약: 관계형 데이터 모델 (계속)

- 기본 갱신 연산 — Insert / Delete / Update(Modify)
- 제약 위반
  - **Insert**: 모든 제약(키·개체·참조) 위반 가능
  - **Delete**: 참조 무결성만
  - **Update**: 키와 참조 무결성
    - PK나 FK 속성을 바꿀 때만 문제

필기 ✍️

★ 시험 포로 외우기 — Insert 3개 / Delete 1개 (참조) / Update 2개(키·참조)

Summary: ER-to-Relational Mapping

❑ Correspondence between ER and Relational Models

ER MODEL	RELATIONAL MODEL
Entity type	*Entity* relation
1:1 or 1:N relationship type	Foreign key (or *relationship* relation)
M:N relationship type	*Relationship* relation and *two* foreign keys
n-ary relationship type	*Relationship* relation and *n* foreign keys
Simple attribute	Attribute
Composite attribute	Set of simple component attributes
Multivalued attribute	Relation and foreign key
Value set	Domain
Key attribute	Primary (or secondary) key

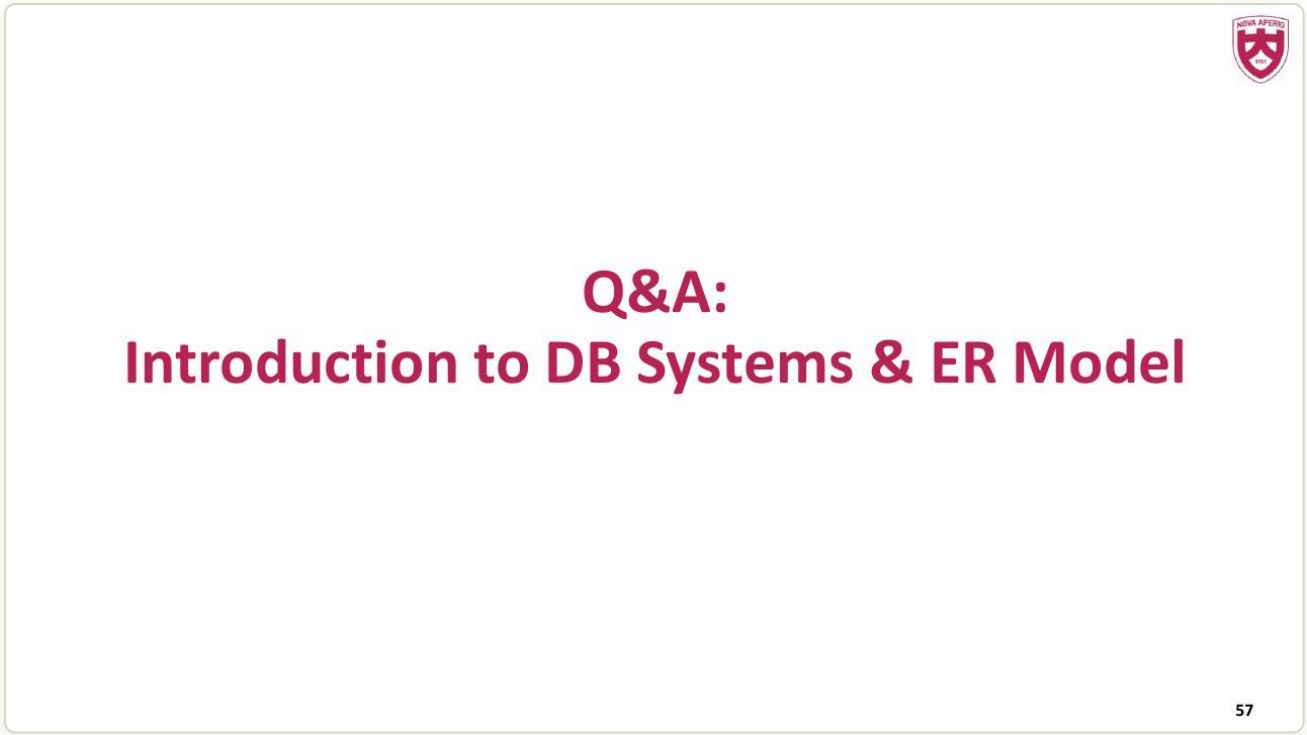
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요약: ER과 관계 모델의 대응

- Entity type → Entity relation
- 1:1, 1:N relationship type → Foreign key (또는 relationship relation)
- M:N relationship type → Relationship relation + FK 2개
- n-ary relationship type → Relationship relation + FK n개
- Simple attribute → Attribute
- Composite attribute → 단순 구성 속성들
- Multivalued attribute → Relation + FK
- Value set → Domain
- Key attribute → Primary (or secondary) key

필기 ✍️

★ 시험 대응표 통째로 — 서술형·빈칸 단골



Q&A

- (슬라이드 제목이 2강 이름으로 남아 있음)

필기 ✍️

💡 이해 뒤쪽 p.58~74는 연습문제 정답

Exercises: Referential Integrity Constraints

❑ Exercise 1

■ Consider the following relations for a database that keeps track of business trips of salespersons in a sales office:

SALESPERSON(Ssn, Name, Start_year, Dept_no)
TRIP(Ssn, From_city, To_city, Departure_date, Return_date, Trip_id)
EXPENSE(Trip_id, Account#, Amount)



■ A trip can be charged to one or more accounts

■ Specify all foreign keys for this schema

Ssn of the TRIP relation references the SALESPERSON relation:
each trip has to be assigned to an existing salesperson

Trip_id of the EXPENSE relation references the TRIP relation:
An expense has to be paid for a particular trip

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연습 1 정답: 외래키

- **TRIP.Ssn → SALESPERSON** — 각 출장은 존재하는 영업 사원에게 배정
- **EXPENSE.Trip_id → TRIP** — 경비는 특정 출장에 대해 지급

필기 ✍️

★ 시험 답 2개 — 화살표 방향(FK → 참조되는 표)까지

Exercises: Referential Integrity Constraints

❑ Exercise 2

■ Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

STUDENT(Ssn, Name, Major, Bdate)
COURSE(Course#, Cname, Dept)
ENROLL(Ssn, Course#, Quarter, Grade)
BOOK_ADOPTION(Course#, Quarter, Book_isbn)
TEXT(Book_isbn, Book_title, Publisher, Author)



■ Specify all foreign keys for this schema

Ssn of the ENROLL relation references the STUDENT relation:
Each enrollment has to include an existing student

Course# of the ENROLL relation references the COURSE relation:
Each enrollment has to include an existing course

Course# of the BOOK_ADOPTION relation references the COURSE relation:
Each book that is being adopted has to belong to a course

Book_isbn of the BOOK_ADOPTION relation references the TEXT relation:
Each book that is being adopted has to have a valid isbn number that can be found

연습 2 정답: 외래키

- **ENROLL.Ssn → STUDENT**
- **ENROLL.Course# → COURSE**
- **BOOK_ADOPTION.Course# → COURSE**
- **BOOK_ADOPTION.Book_isbn → TEXT**

필기 ✍️

⚠ 주의 Quarter는 FK 아님 — 참조할 표가 없다

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

a. Insert <'Robert', 'F', 'Scott', '943775543', '1972-06-21', '2365 Newcastle Rd, Bellaire, TX', M, 58000, '888665555', 1> into EMPLOYEE.

All constraints are satisfied

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3 정답 (a)

- Insert <'Robert', 'F', 'Scott', '943775543', ..., '888665555', 1> into EMPLOYEE
- → **모든 제약 만족 — Super_ssn 888665555, Dno 1 모두 존재**

필기 ✍️

✏ 예.풀이 (a) 모든 제약 만족 — Super_ssn 888665555, Dno 1 모두 존재

💡 이해 확인 순서: PK 중복 → PK NULL → FK 대상 존재 → (삭제면) 나를 가리키는 행

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

b. Insert <'ProductA' 4, 'Bellaire' 2> into PROJECT.

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

Referential IC is violated

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3 정답 (b)

- Insert <'ProductA', 4, 'Bellaire', 2> into PROJECT
- → **참조 무결성 위반 — Dnum=2인 부서가 없음**

필기 ✍️

✏ 예.풀이 (b) 참조 무결성 위반 — Dnum=2인 부서가 없음

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

c. Insert <'Production' 4 '943775543', '2007-10-01'> into DEPARTMENT.

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

The key constraint is violated

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3 정답 (c)

- Insert <'Production', 4, '943775543', '2007-10-01'> into DEPARTMENT
- → **키 제약 위반 — Dnumber 4가 이미 있음 (슬라이드 답)**

필기 ✍️

✏ 예.풀이 (c) 키 제약 위반 — Dnumber 4가 이미 있음 (슬라이드 답)

⚠ 주의 엄밀히는 Mgr_SSN 943775543인 직원도 없어서 참조 무결성도 어긋난다(이 연습은 원래 상태에 바로 적용). 슬라이드 답은 키 제약만 — 서술형이면 둘 다 쓰고 근거를 달자

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

d. Insert <'677678989', NULL, '40.0'> into WORKS_ON.

The referential and entity IC are violated

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3 정답 (d)

- Insert <'677678989', NULL, '40.0'> into WORKS_ON
- → **개체 무결성(Pno는 PK의 일부인데 NULL) + 참조 무결성(Essn 677678989 직원 없음) 위반**

필기 ✍️

★ 시험 (d) 개체 무결성(Pno는 PK의 일부인데 NULL) + 참조 무결성(Essn 677678989 직원 없음) 위반

⚠ 주의 위반이 두 개 — '모두' 적으라는 문제에서 하나만 쓰면 감점

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

e. Insert <'453453453', 'John' 'M', '1990-12-12', 'spouse'> into DEPENDENT.

No IC is violated

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3 정답 (e)

- Insert <'453453453', 'John', 'M', '1990-12-12', 'spouse'> into DEPENDENT
- → **위반 없음**

필기 ✍️

✏ 예.풀이 (e) 위반 없음

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

f. Delete the WORKS_ON tuples with Essn = '333445555'

No IC is violated

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3 정답 (f)

- Delete the WORKS_ON tuples with Essn = '333445555'
- → **위반 없음 — WORKS_ON 튜플을 참조하는 곳이 없음**

필기 ✍️

✏ 예.풀이 (f) 위반 없음 — WORKS_ON 튜플을 참조하는 곳이 없음

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

g. Delete the EMPLOYEE tuple with Ssn = '987654321'.

Referential IC is violated

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3 정답 (g)

- Delete the EMPLOYEE tuple with Ssn = '987654321'
- → **참조 무결성 위반 — WORKS_ON, DEPENDENT, DEPARTMENT.Mgr_ssn, EMPLOYEE.Super_ssn이 참조**

필기 ✍️

✏ 예.풀이 (g) 참조 무결성 위반 — WORKS_ON, DEPENDENT, DEPARTMENT.Mgr_ssn, EMPLOYEE.Super_ssn이 참조

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

h. Delete the PROJECT tuple with Pname = 'ProductX'.

Referential IC is violated

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3 정답 (h)

- Delete the PROJECT tuple with Pname = 'ProductX'
- → **참조 무결성 위반 — WORKS_ON의 Pno=1 튜플들이 참조**

필기 ✍️

✏ 예.풀이 (h) 참조 무결성 위반 — WORKS_ON의 Pno=1 튜플들이 참조

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

i. Modify the Mgr_ssn and Mgr_start_date of the DEPARTMENT tuple with Dnumber = 5 to '123456789' and '2007-10-01', respectively.

No IC is violated

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3 정답 (i)

- Modify Mgr_ssn, Mgr_start_date of DEPARTMENT Dnumber=5 to '123456789', '2007-10-01'
- → **위반 없음 — 123456789 직원이 존재**

필기 ✍️

✏ 예.풀이 (i) 위반 없음 — 123456789 직원이 존재

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

j. Modify the Super_ssn attribute of the EMPLOYEE tuple with Ssn = '999887777' to '943775543'.

Referential IC is violated

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3 정답 (j)

- Modify Super_ssn of EMPLOYEE Ssn='999887777' to '943775543'
- → **참조 무결성 위반 — 943775543 직원이 없음**

필기 ✍️

✏ 예.풀이 (j) 참조 무결성 위반 — 943775543 직원이 없음

Exercises: Constraint Violation (Cont.)

❑ Exercise 3

k. Modify the Hours attribute of the WORKS_ON tuple with Essn = '999887777' and Pno = 10 to '5.0'.

No IC is violated

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

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연습 3 정답 (k)

- Modify Hours of WORKS_ON Essn='999887777' AND Pno=10 to '5.0'
- → **위반 없음 — PK·FK가 아닌 속성**

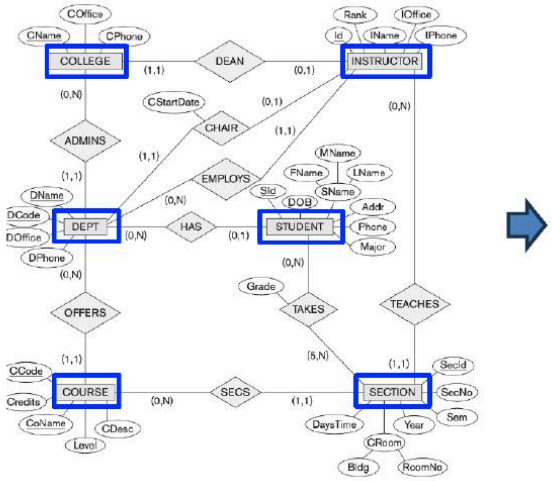
필기 ✍️

✏ 예.풀이 (k) 위반 없음 — PK·FK가 아닌 속성

Exercises: ER-to-Relational Mapping

❑ Exercise 4

■ Step 1: Mapping of Regular Entity Types



COLLEGE

CName	COffice	CPhone

INSTRUCTOR

Id	Rank	IName	IOffice	IPhone

DEPARTMENT

DName	DCode	DOffice	DPhone

COURSE

CCode	Credits	CoName	Level	CDesc

SECTION

SecId	SecNo	Sem	Year	Bldg	RoomNo	DaysTime

STUDENT

Sid	DOB	FName	MName	LName	Addr	Phone	Major

/1

연습 4 정답 — Step 1: 일반 엔티티

- COLLEGE(**CName**, COffice, CPhone)
- INSTRUCTOR(**Id**, Rank, IName, IOffice, IPhone)
- DEPARTMENT(**DName**, DCode, DOffice, DPhone)
- COURSE(**CCode**, Credits, CoName, Level, CDesc)
- SECTION(**SecId**, SecNo, Sem, Year, Bldg, RoomNo, DaysTime)
- STUDENT(**Sid**, DOB, FName, MName, LName, Addr, Phone, Major)

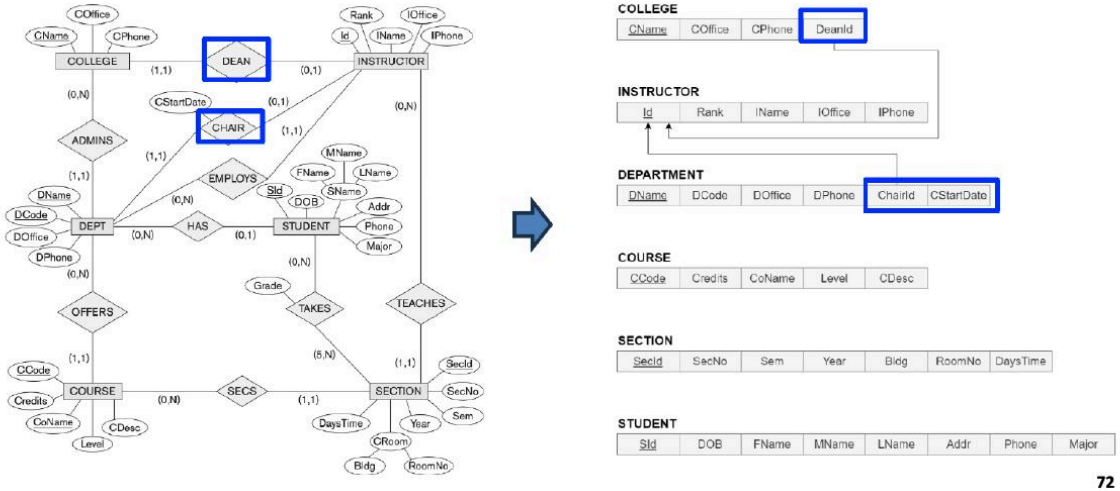
필기 ✍️

💡 이해 STUDENT의 Name(FName, MName, LName)은 복합 ⟶ 세 칸으로

Exercises: ER-to-Relational Mapping

❑ Exercise 4

■ Step 2: Mapping of Binary Relationship Types (1:1)



연습 4 정답 — Step 2: 1:1

- DEAN → COLLEGE에 **DeanId** (FK → INSTRUCTOR)
- CHAIR → DEPARTMENT에 **ChairId, CStartDate** (FK → INSTRUCTOR, 관계 속성 포함)

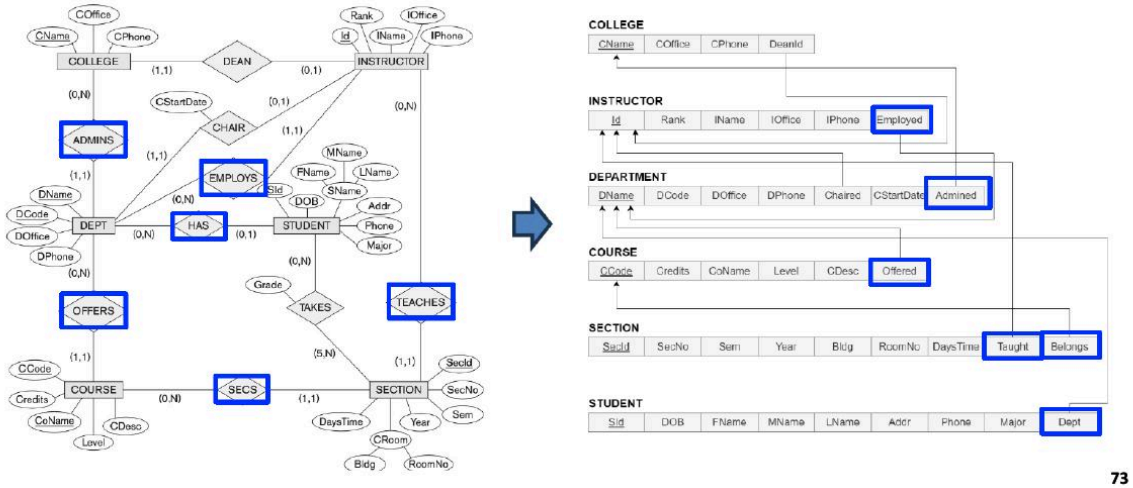
필기 ✍️

★ 시험 1:1은 total 쪽(COLLEGE, DEPARTMENT)에 FK

Exercises: ER-to-Relational Mapping

❑ Exercise 4

■ Step 3: Mapping of Binary Relationship Types (1:N)



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연습 4 정답 — Step 3: 1:N

- EMPLOYS → INSTRUCTOR에 **Employed** (DEPT 참조)
- ADMINS → DEPARTMENT에 **Admined** (COLLEGE 참조)
- OFFERS → COURSE에 **Offered** (DEPT 참조)
- TEACHES → SECTION에 **Taught** (INSTRUCTOR 참조)
- SECS → SECTION에 **Belongs** (COURSE 참조)
- HAS → STUDENT에 **Dept** (DEPT 참조)

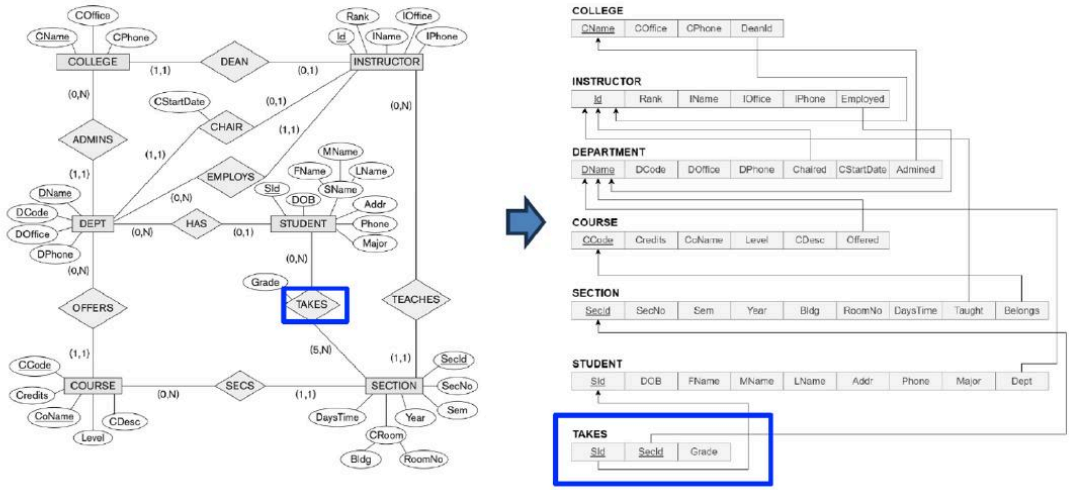
필기 ✍️

★ 시험 1:N은 언제나 N쪽 표에 FK

Exercises: ER-to-Relational Mapping

❑ Exercise 4

■ Step 4: Mapping of Binary Relationship Types (M:N)



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연습 4 정답 — Step 4: M:N

- TAKES → 새 릴레이션 **TAKES(SId, SecId, Grade)**
  - PK = (SId, SecId), 관계 속성 Grade

필기 ✍️

★ 시험 M:N 하나(TAKES)만 새 표가 생긴다